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Continuing Commentary

Commentary on **Edward P. Stabler, Jr. (1983) How are grammars represented? BBS 6:391–421.**

Abstract of the original article: Noam Chomsky and other linguists and psychologists have suggested that human linguistic behavior is somehow governed by a mental representation of a transformational grammar. Challenges to this controversial claim have often been met by invoking an explicitly computational perspective: It makes perfect sense to suppose that a grammar could be represented in the memory of a computational device and that this grammar could govern the device's use of a language. This paper urges, however, that the claim that humans are such a device is unsupported and that it seems unlikely that linguists and psychologists really want to claim any such thing. Evidence for the linguists' original claim is drawn from three main sources: the explanation of language comprehension and other linguistic abilities; evidence for formal properties of the rules of the grammar; and the explanation of language acquisition. It is argued in this paper that none of these sources provides support for the view that the grammar governs language processing in something like the way a program governs the operation of a programmed machine. The computational approach, on the contrary, suggests ways in which linguistic abilities can be explained without the attribution of any explicit representation of rules governing linguistic behavior.

Language processing and computer programs

R. J. Harvey
Department of Anatomy, University of Otago Medical School, Dunedin, New Zealand

For language to be a practical form of communication, language processing must be rather consistent and must therefore depend on rules of some sort, although they do not necessarily form a "transformational grammar" such as Chomsky (e.g., 1980) has proposed. [See Chomsky: "Rules and Representations," *BBS* 3(1) 1980.] For humans, language processing depends on the brain. The great majority of neuroscientists would accept that the operations of the brain result from the interactions of its individual neurones according to a set of consistent rules. If so, it would be possible, in principle, to construct a computer program (or other system) which would simulate the activities of the brain. Such a program would need to incorporate not only the rules by which the neurons interact (including rules which generate memory and other "plastic" properties of the brain), but also details of the interconnections between the neurones. In our present state of knowledge, we can sketch only the faintest outlines of such a program and, with existing general-purpose computers, it would run many orders of magnitude slower than the real brain. (A program which gives a reasonable simulation of the activity of just a single, simplified neurone has been found to run slower than "real" time [e.g., Harvey 1978].)

To interact with the outside world, a brain requires transducers and to convert patterns of activity in motoneurones into movements and to convert afferent input into patterns of neural activity; these patterns are symbols representing the inputs. For language processing, the original input is therefore in terms of symbols for symbols, and these are processed in the brain to produce more symbols – perhaps at this level representing, for written input, the letters (or groups of letters) of the message. Further processing may produce symbols represnting words.

To respond appropriately to a complete message, additional levels of processing will clearly be required. (In practice, of course, the processing does not proceed only from lower to higher levels; the results of lower level manipulations may be modified in the light of higher level processing.) At each level, the manipulations must depend ultimately on the neurones in the brain, but it is possible that the rules of symbol manipulation could be expressed directly, rather than in terms of the activities of the neurones. A program simulating the manipulations directly would almost certainly require a great deal less computation than one simulating the neurones individually. (To take a very simple example, it is easy to think of a symbol manipulation algorithm to extract words from a printed message, provided thewordsarenotruntogether – although this is clearly not the only such algorithm available to the brain!) Higher level symbol manipulation rules must depend on lower level ones, but it may be possible to describe or define them without explicit reference to the lower level rules. (In much the same way, it is possible to describe the physical properties of water without explicit reference to the atoms which make it up.)

When Chomsky (e.g., 1980) describes language processing as being "rule-governed," these rules must be at a level comparable to the high level ones described above. The rules differ from person to person depending on which particular language(s) they know, but the low level rules should be much the same for all humans, because their brains are neuroanatomically and neurophysiologically very similar. Similarity of the lower level rules would explain why there are parallels between the processing rules for different languages. Moreover, Chomsky's (and others') success in finding elements of a "grammar" supports the idea that the rules can be expressed in terms which do not explicitly incorporate lower level rules and that they are computable at some level, making them consistent with Chomsky's (1980) "computational perspective." However, the writing of a program to carry out the operations has proved very difficult, partly at least because currently available computers, computer languages, and programming techniques have been designed for a very different type of processing. Even if the rules at one level could be defined, it would not necessarily allow the rules for a different level to be discovered; language processing rules derived from linguistic studies have not given any real clue as to the neurophysiological and neuroanatomical substrates for them – but that does not mean that those substrates are not there.

I do not believe that the above is unduly reductionist. Properties of a system which are apparent at a particular level of description may well be implicit in lower level properties, but cannot necessarily be deduced from a knowledge of them. Even if higher level properties can in some sense be explained by the lower level ones, this does not "reduce" them to a lower level, nor does it explain them away (see also Miller 1981).

When running, a computer program simulating a physical

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system manipulates symbols representing the attributes of elements of the system, and although its operations may be analogous, they are clearly somewhat different from those of the real system. However, for most levels of processing, the brain also manipulates symbols (or symbols representing symbols, and so on), and the same sort of distinction between a simulation and the real thing may not apply. A brain simulation, when operating, may well possess internal states similar to those of a brain; these could include what, for brains, are known as mental states.

The role of a brain is very different from that of a general-purpose computer which is running a brain-simulation program; the parallels between computers and brains which have sometimes been invoked are misleading (see also Gross 1983). When it is running a brain-simulation program, a computer is not part of the system; it bears a relationship to it roughly equivalent to that of oxidative metabolism to the brain – a necessary basis, but not related to its specialised brainlike functions. (Indeed, the program would produce the same results however it was implemented; an electronic computer is merely the fastest currently available device. Similarly, a brain would work equally well if its cells could be supplied with energy in an alternative way.)

Stabler (1983), in his target article and response, appears not to have made the correct distinction between the role of a brain and that of a computer running a brain-simulation program. Most of the information incorporated in a "bottom-level" program would relate to the interconnections of the neurones of the brain, whereas for higher level programs, it would relate to the symbol manipulation rules. For all programs, as a number of commentators have said, whether such information is called "data" or "program" is irrelevant, and there seems no reason why a "mental" representation should not be in the same category. The distinctions which Stabler draws between the M and the various versions of the H hypotheses are similarly irrelevant. They represent different ways of implementing the system but do not affect what it does; they are at a different level of description.

Many commentators have made the point that a device (or program) which processes data according to a set of rules is a representation of those rules. Stabler (1983) implies that something more is needed; that is, that the rules must be contained in the system in some form additional to this in order for it to be considered a representation. A good deal of the debate stems from different interpretations of what is meant by "representation." It does not seem to me that Stabler has shown that his additional connotation of "representation" has any functional significance – although it could be useful as a metaphor.

Author's Response

Computational models of language processing

Edward P. Stabler, Jr.

Quintus Computer Systems Inc., Palo Alto, CA 94306

Most of Harvey's remarks are in accord with the point of view defended in the 1983 target article: He notes that there is certainly a consistency among the language-processing mechanisms of different people; that these mechanisms may be describable at a number of levels as computational systems in which symbols are manipulated; that a computer might also manipulate symbols in something like the way the brain does and thereby come to be in states which are similar to brain states; and that some of these states of symbol manipulating systems like brains and computers might in fact be mental states. (Beside these points of agreement on matters of great controversy, the points of contention seem almost insignificant!) The disagreement comes only with Harvey's claim that, contrary to my argument, Chomsky and other linguists can be seen as providing well-motivated accounts of the rules that are represented and used in language processing, and that my distinctions between various representational claims are irrelevant.

The argument of the target article can be summed up as follows. Many linguists and not a few psychologists have argued that grammars of a certain kind (e.g., transformational grammars) are mentally represented and used in the exercise of linguistic abilities, much as a grammar might be used by a computer. A grammar for an individual's language might be "represented and used" in the exercise of linguistic abilities either as a "program" which is executed or as "data" which are accessed in language understanding and use. These two possibilities were referred to in the target article as hypotheses (H2) and (Hd), respectively. The evidence for the representational claims made by linguists and psychologists was reviewed, and it was shown that this evidence supported at most the view that rules of a grammar may be *computed* in the exercise of linguistic abilities. This last, more modest, hypothesis, which makes no representational claims, was referred to as (H2). Because a rule can be computed without being represented (in the sense in which some simple calculators can be said to compute rules for finding sums without accessing any data or program that represent those rules), the evidence does not support the representational claims of *either* (H2) *or* (Hd). Furthermore, machines (like simple calculators) that do not access representations of programs that we describe them as computing are, *ceteris paribus*, faster and simpler than machines that do access representations (like Turing machines or programmable computers), and so the less ambitious hypothesis (H2) should be preferred to the representational claims. So even if we grant that the rules of grammar in some sense characterize the functions being computed in language understanding, there is still no reason to accept the stronger representational claims about how those computations are carried out.

This argument has not been undermined by any evidence since the target article was written. This is not surprising because, as I pointed out in the target article, the evidence for representational claims would have to come from theories about how linguistic processes were actually implemented, and we do not seem to be very near to such an understanding. Furthermore, I have argued (Stabler 1984) that there is no good reason to think that grammars like Chomsky's even characterize the functions being computed in language understanding, let alone provide the rules represented and used in those computations. The grammars might have some subtle but still substantial relation to the functions computed, but the relation remains obscure.

Harvey says that I have misunderstood the distinction between the role of a brain and the role of a general-purpose computer when it is running a brain-simulation program. It is not quite clear what distinction Harvey has in mind here. He points out that a brain-simulation

program would produce the same results however it was implemented, just as a brain might work in the same way if it got most of its energy from something other than oxidative metabolism, but the relevant contrast between brains and computers is not made quite clear by this point. Is the point just that a brain is not a general-purpose computer? Perhaps the brain is not general-purpose (in the sense that there are some computable functions that it cannot compute), but nothing in the target article assumes otherwise. As Harnish (1983) pointed out in his commentary, a computer could be built so that it would execute only one program, a program which is accessed and used in the way it would be accessed by a general-purpose computer. (As Harnish says, unused portions of the system may be "blown out.") So even if the brain is not "general-purpose," it could still represent and use programs in the way a computer can. All of this is perfectly compatible with the account of computation offered in the target article.

Another thing that Harvey might be worried about is the idea that the computer, unlike the brain, is running a brain-simulation program. This is a reasonable point. Notice that nothing in the target article talked about "brain-simulation" programs. What is contemplated is the possibility that we will be able to account for certain state transformations that occur in the brain in symbolic terms, just as we can account for certain state transformations that occur in certain electric circuits (e.g., computers) in symbolic terms. In other words, a computer and a brain might compute the same functions or programs, under appropriate interpretations of their states. So for example, in visual perception, the brain might compute a function from the array of light frequencies and intensities at the retina to a "primal sketch" of two-dimensional relationships, and then a function from those primitive representations to representations of three-dimensional relationships, as Marr (1982) and others have proposed. If we could characterize these functions completely enough, we could program a computer to compute those very same functions. Neither the brain nor this programmed computer would be computing a "brain-simulation program" in any natural sense. Rather, we would have two different kinds of systems, brains and computers, computing the same program. At this level of description, brains and computers do have the same role: They are both just physical systems whose state transformations correspond to certain functions (under an appropriate interpretation of states, as explained in the target article).

Harvey says that the distinctions between the various hypotheses considered in the target article "represent different ways of implementing the system but do not affect what it does," which is exactly the view of the target article. But Harvey then says that these distinctions are "irrelevant," without ever quite saying what they are irrelevant *to*. Certainly, on his own account (as just quoted), they are not irrelevant to theories about how a process is implemented. and that was exactly the claim of the target article. The representational claims are concerned with matters of implementation about which there is little evidence. And among the possible views about implementation, the representational hypotheses are not the strongest.

So perhaps the only real disagreement between Harvey and me is that he agrees with those commentators who said that "a device (or program) which processes data according to a set of rules is a representation of those rules." This proposal just cannot make sense of the issues that the target article was addressing. Consider any system that processes data according to a set of rules. According to Harvey's view, this system is a representation of those rules. But then consider any other sets of rules which describe the same processing of data; by the same reasoning, the system is a representation of each of those sets of rules as well! In other words, representing a set of rules, on this view, comes to nothing more than behaving in accordance with those rules. This view is unable to make sense of the controversy about whether the performance of certain tasks merely conforms to certain rules or whether it actually involves the representation and use of rules. There *are* differences between various sets of rules that define the same processing of data, and sometimes it is appropriate to describe a system as using a representation of one and not another of these sets of rules. There are numerous results in computer science about the relative virtues of different sets of rules for the same computation (e.g., some sets of rules contain more symbols than others; some contain many symbols, but in a sequence that is easily defined by another program, etc.), and there is no reason to think that we could not speak in the same sense about the properties of some particular rules which are represented and used by the brain. All of this would be nonsense on the view Harvey proposes, according to which every system that represents and computes one set of rules represents every set of rules that defines the same computation. The point of the target article, though, is that we have no reason to think that an account of represented rules is appropriate in the case of human language processing, though our understanding of the relevant aspects of human language processing and their implementation is presently at a very early stage.

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Commentary on **Donald E. Broadbent (1984) The Maltese cross: A new simplistic model for memory. BBS 7:55–94.**

Abstract of the original article: This paper puts forward a general framework for thought about human information processing. It is intended to avoid some of the problems of pipeline or stage models of function. At the same time it avoids the snare of supposing a welter of indefinitely many separate processes. The approach is not particularly original, but rather represents the common elements or presuppositions in a number of modern theories. These presuppositions are not usually explicit, however, and making them so reduces the danger of slipping back into earlier modes of thought.

The key point is to distinguish between persisting representations and the processes that translate one representation into another. Various classes or groups of persisting representations can be distinguished by the experimental treatments that interfere with them. In particular, there now seem to be several kinds of short-term or temporary storage, different from each other as well as from long-term memory; the translating processes also have several different modes or kinds. A particularly important aspect of the current position is that a model of this general type no longer requires some external agent to direct and control long sequences of behaviour.

Putting the Maltese cross into context

A. J. R. Doyle

Department of Psychology, University of Sheffield, Sheffield S10 2TN, England.

The exchange of views on the Maltese cross model in *The Behavioral and Brain Sciences* (Broadbent 1984a) failed to differentiate between two critically different modes of operation for the brain.

Simple systems. 1.1. Consider a simple organism such as the Marine Snail Tritonia. A drop of concentrated salt anywhere on its body elicits an alternating sequence of contractions of the ventral and dorsal muscle groups producing a typical escape manoeuvre.

1.2. Electromechanical model. Such a system could be modelled from electromechanical components. The skin surface could have contacts which operate a switch to bring the muscle contractions into action. These could be modelled with some sort of pulse generator that alternatively operated solenoids.

1.3. Symbolic model. It would be possible to replace the detector switch and the pulse generator with a computer, so that the current from the simulated skin sensor was the input and the sequence of voltages to the simulated muscles the output.

1.4. Identity by spatial location. The critical difference between these two models is that in the electronic version the identity of the voltage values is determined by the physical location of the wires, whereas in the computer version both the value and identification of any event are symbolised by the pattern of the various memory switches. Where this pattern actually exists physically is not dictated by the program, nor is it necessarily the same each time the program is run. The voltages running through the on/of switches of the machine have no fixed relation to the meaning attributed to them in the program.

1.5. The critical difference. The difference between the models leads to a critical difference in their use. In the electromechanical model a detailed observation of its operation would lead to a complete wiring diagram, and equally a complete wiring diagram would lead to an exact knowledge of its operation. With the computer model an examination of the physical components could never lead to a knowledge of the way it operated, and although a detailed observation of its performance would lead to proposals for possible programs, the actual program is absolutely necessary before one can tell which system it is actually using.

Human behavior – the essential question. 2.1. In a simple system of this sort it is easy to propose an electromechanical model. If the system itself used identification by symbol instead of physical location, however, it would be impossible to model it in this way. The essential question is whether the human brain can carry out its tasks using a system in which the states that represent events in the external world are identified by their physical position.

2.2. The rationale. A system that uses identification by physical position seems unlikely because the number of events is enormous. The argument for identification by symbolic labelling is that it enables the system to use a limited number of specialised physical devices for completely different operations.

The value of the Maltese cross. 3.1. The value of the Maltese cross depends on which view one takes of the way the brain works.

3.2. Identification by location. Identification by location presupposes that it is possible to relate each physical detail to some function expressed in input and output terms. The cross would then represent the first essential step in describing the input/output aspects of the system. From this perspective the success of its appeal for generality depends on the extent to which performance and structure can be considered under class headings and still preserve the one-to-one mapping inherent in such a system. Such an approach depends upon there being only minor differences in detail between individuals.

3.3. Identification by label. If it is supposed that at some stage the system uses identification by label employing hierarchically organised layers of virtual machines supported by an underlying physical machine, then the situation is quite different. No amount of research into performance will lead to constraints on the physical construction of the underlying mechanism. Similarly, all details of the physical function in terms of storage capacity, power consumption, and the equivalent of register organisation could be known without knowing anything of the programs running on it. Consequently any reference to the physical organisation of memory would be inappropriate.

Although knowing the input and output at the symbolic level allows algorithms to be proposed which might produce similar transformations, nothing can be done to show what has actually happened without knowing the complete sequence of patterns in the brain of a particular individual and what it represents.

Conclusion. Any model of brain and behaviour, and criticisms of such models, should be preceded by a statement about which of the two systems is assumed to underlie the data because of the critical difference it makes to the interpretation. Failure to make this point explicit leaves one permanently hovering on the threshold of an explanation that never seems to materialize.

Organization of long-term and working memory stores

Gregory V. Jones

Department of Psychology, University of Bristol, Bristol BS8 1HH, England

Broadbent (1984a) provides a coherent synthesis of recent developments in the information-processing approach to cognition. Whereas most recent work on memory has concerned itself with investigating specialized components, Broadbent takes a wider approach to the system as a whole. The analysis is

concerned primarily not with the architecture of individual structures, but instead with the interaction between structures – adopting the scale of the town planner rather than that of the architect, perhaps. It is clear that the two approaches are complementary rather than competing, and that there still remains a formidable amount to learn about each component of Broadbent's analysis. Probably nowhere is this truer than in the case of the long-term associative store.

A considerable amount of evidence indicates that, as Broadbent suggests, a basic format in which event co-occurrence information is stored is that of the fragment (e.g., Jones 1976; 1978). Each memory fragment contains a limited set of event representations whose members can be shown (by cued recall experiments) to be all directly accessible to each other. Further evidence – in particular, that reported by Ross and Bower (1981) – seems to demonstrate, however, that a second and distinct unit of memory organization is also identifiable. The members of a set of event representations may alternatively be linked to each other indirectly rather than directly, by virtue of their relations with a common mediating schema (Jones 1983b; 1984). According to this view, then, the organization of the long-term associative store possesses a degree of complexity beyond that outlined by Broadbent. However, as noted by Morton (1984), modification of the detailed format of organization of the associative store in this way does not undermine the overall appropriateness of the Maltese cross model.

Broadbent points out also that the long-term store must be capable of registering the overall probabilities with which different events co-occur over relatively extensive time spans. It has been convincingly argued by Rosch (1978) that it is in response to these contingencies that our category concepts themselves are developed. Certain categories are basic in that stimuli are processes most efficiently as members of those categories (see Mervis & Rosch 1981; Murphy & Smith 1982). Basic categories appear to be identifiable via the criterion of occurring at "the most general and inclusive level at which categories can delineate real-world correlational structures" (Rosch, Mervis, Gray, Johnson & Boyes-Braem 1976, p. 384). The question of determining the particular aspect of event co-occurrence contingencies to which the long-term associative memory is sensitive in the formation of natural categories may therefore be approached by attempting to refine this criterion. Rosch and her colleagues have proposed that the relevant criterion is that of cue validity: the extent to which a given feature of the environment is a reliable cue for the different categories under comparison. Some recent work suggests, however, that the relevant factor is the conjunction of both cue validity and its complement (Jones 1983a). That is, category basicness derives not only from the extent to which feature possession predicts category membership, but also from the extent to which category membership predicts feature possession. This general area of analysis is of particular interest in the present context because it could also be described as concerned with the induction of generic or semantic memory from individual or episodic memories – for a detailed consideration of this distinction, see the BBS multiple book review of Tulving's *Elements of episodic memory*, *BBS* 7(2) 1984 – and perhaps illustrates that the long-term associative store may encompass not only the latter but also the former.

Let us revert now to viewing the plan of cognition drawn by Broadbent from a global rather than a local perspective. The configuration of three of the four limbs of the Maltese cross – the left-hand and right-hand segments that relate to receptors and effectors, respectively, and the bottom segment that constitutes the long-term associative store – seems incontrovertible. In principle less secure, as noted by Schneider (1984) and by Broadbent himself, is the top segment that contains the abstract working memory. So could the Maltese cross be truncated into a tau cross? Broadbent gives cogent reasons for the fourth limb's retention. In particular, he points to its role in temporarily holding information that is needed to maintain control over the system's operations, and notes that this may be explored further by investigating instances of cognitive failure.

The incidence with which cognitive failures are reported to occur in ordinary life has been assessed by Broadbent, Cooper, Fitzgerald, and Parkes (1982), and in the present context may usefully be related to individuals' performance on laboratory tasks. In reviewing recent work on the relation between everyday cognitive failure and several different aspects of laboratory performance, Martin and Jones (1984) found little evidence of the effects of two possible factors. First, the effect of irrelevant processing during the performance of a laboratory task may be studied using focussed-attention paradigms such as the Stroop method (Stroop 1935), but does not itself appear to be important. Second, the effects upon performance of simple memory load also do not seem to be related to everyday cognitive failure. In contrast, a third factor does appear to be of significance. This is the effect of that distribution of attention which is necessary in order to carry out two unpredictable tasks concurrently (e.g., to both read a text and detect auditory targets simultaneously). The preceding three factors correspond approximately to the three successive construals of the way in which control information is held temporarily that were considered by Broadbent when discussing the question of the central executive. Thus on present evidence his third interpretation may be favoured. This asserts that control information is held temporarily in a special-purpose component of the abstract working memory, susceptible to interference only from other control information. A dedicated substore of this type would be consistent with recent suggestions (Allport 1980; Martin & Jones 1983) of the existence of a mechanism that is devoted to keeping track of multiple concurrent task goals.

Author's Response

The computation of control

Donald E. Broadbent

Department of Experimental Psychology, University of Oxford, Oxford OX1 3UD, England

The commentaries by **Jones** and by **Doyle** raise interesting matters to which any response must be further elaboration of the points raised, not contradiction. As Jones indicates, the structure of long-term memory was handled only briefly in the target article; the approach to which he refers, through the structure of natural categories, and through the interplay between episodic and semantic records, is indeed an exciting and hopeful one. I would add one gloss; new contingencies can apparently be learned in either an explicit fashion, reportable by the person, or in an implicit one (Berry & Broadbent 1984; Reber, Kassin, Lewis & Cantor 1980). The distinction may have to do with the route followed by the information from the outer world to the long-term store. It could travel indirectly and selectively by way of one of the temporary stores or else directly, unselectively, and therefore building up the contingency only slowly (see the discussions of "Flint's Conjecture" by Broadbent 1981; 1984b).

On the question of control, like **Jones** I find that my own (unpublished) data suggest a connection between reported cognitive failure and the ability to handle separate sources of information. Let us be cautious for a while, however, in interpreting these results, because they

seem to change or reverse depending on the current anxiety level of the person. The pioneering work of Martin and Jones (1983; 1984) is based on small and rather highly selected samples that may not have been typical. At least there seems no sign, either in our data or in those of Martin and Jones, that cognitive failure is associated with a deficiency of temporary memory, or with a deficiency of total processing. It has something to do with selectivity.

When this matter is clarified, we shall have an understanding at the computational level of Marr (1982), not at his level of implementation. That is, we shall know what operations the nervous system carries out, but in no sense where or what the nervous machinery is that undertakes those operations. In terms of **Doyle's** distinction, we should have a symbolic model but not an electromechanical one. No purely psychological studies can achieve understanding at the level of implementation; just as no observation in vitro of events at one location in the nervous system can establish understanding at the computational level. It scarcely seems necessary to preface every contribution to brain science by a statement of the level it is discussing, because these restrictions apply to all of them. In case of any ambiguity, however, the Maltese cross is a model at the computational level.

Two glosses are worth adding. First, it is essential to establish the correct computational ("symbolic") model in order to verify a model of implementation. If an electromechanical model of Tritonia was not describable by the same symbolic model that correctly fitted the behavior of the snail, then the electromechanical model would be false. Examples already exist of such interaction between levels; for instance, a simple theory of the working of the ear might have been that it performs a perfect Fourier analysis on the acoustic input, and the events at various locations in the auditory pathway could then be interpreted to be consistent with that view. In fact, however, we know from the behavior of people listening to complex combinations of tones that the ear is doing a rather different computation, and theories of the physiological mechanism must be constrained by those results (Leakey, Sayers, & Cherry 1958; Moore 1973). Similarly, the insights of Marr (1982) into the computations performed by the visual system change our interpretation of results found at the physiological level.

The second gloss concerns explanation in terms of location. One of the attractions of simple nervous systems, such as those of invertebrates, is that one might hope to relate a particular output to activity at a particular location in the system [see Hoyle: "The Scope of Neuroethology," *BBS* 7(3) 1984]. In complex systems, however, the possibility arises that particular events may be represented by distributed coding; no event at a single location corresponds to a single output, and any single location corresponds to a wide variety of different outputs depending on events at other locations. Such an arrangement is likely on general engineering grounds (Broadbent 1985); there is also a good deal of evidence that it does occur in mammalian nervous systems. For example, in hearing, information on the temporal periodicity of the acoustic input is carried by the whole nerve when the frequencies concerned are sufficiently high that no single fiber could be responding to every peak of pressure (Moore 1973).

Accordingly, even at the level of implementation, explanations in terms of location are likely to be possible only "in principle." To find true explanations through location would require observation of every location simultaneously. The construction of broad computational or symbolic models, such as the Maltese cross, is likely to avoid misinterpretation of observations at individual locations.

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Commentary on **Michael E. Lamb, Ross A. Thompson, William P. Gardner, Eric L. Charnov, and David Estes (1984) Security of infantile attachment as assessed in the "strange situation": Its study and biological interpretation. BBS 7:127–71.**

Abstract of the original article: The Strange Situation procedure was developed by Ainsworth two decades ago as a means of assessing the security of infant-parent attachment. Users of the procedure claim that it provides a way of determining whether the infant has developed species-appropriate adaptive behavior as a result of rearing in an evolutionarily appropriate context, characterized by a sensitively responsive parent. Only when the parent behaves in the sensitive, species-appropriate fashion is the baby said to behave in the adaptive or secure fashion. Furthermore, when infants are observed repeatedly in the Strange Situation, the pattern of behavior is said to be highly similar, and this pattern is said to predict the infants' future behavior in a diverse array of contexts. After an exhaustive review of the literature, it is shown that these popular claims are empirically unsupported in their strong form, and that the interpretations in terms of biological adaptation are misguided. There is little reliable evidence about the specific dimensions of parental behavior that affect Strange Situation behavior, although there does appear to be some relationship between these constructs. Temporal stability in security of attachment is high only when there is stability in family and caretaking circumstances. Likewise, patterns of Strange Situation behavior only have substantial predictive validity in similarly stable families. Implications for future research and theorizing – particularly as they relate to the use of evolutionary biology in psychological theory – are discussed.

The development and component processes of the attachment system: Some suggestions for their rediscovery

J. P. Connell

Graduate School of Education and Human Development, University of Rochester, Rochester, N.Y. 14627

At the heart of the Lamb, Thompson, Gardner, Charnov, and Estes (1984) review is the proposition that most of the hypotheses put forth by Ainsworth and her colleagues regarding the ontogenetic and phylogenetic significance of individual differences in the security of infantile attachment are not currently supported by the available empirical evidence. As these authors and others before them have pointed out, the empirical studies being brought to bear on the hypotheses of "traditional" attachment theorists all share a three-group typological summary of individual differences in security of attachment (the ABC classification system) that is drawn from a 20-minute laboratory procedure (the Strange Situation). These methodological features of previous attachment work are cited by Lamb et al. and some of the commentators as being partially responsible for what they see as the weak empirical foundation of the strong theoretical claims made by attachment theorists such as Ainsworth and Sroufe.

The authors of the target article argue quite convincingly that by employing the ABC typology as the sole indicator of security of attachment some crucial variability in patterns of attachment behavior is obscured and that this problem is exacerbated when the two insecure types are collapsed into one group. Further, the authors (and, more cogently, the commentators) take the traditional attachment research to task for relying so heavily on a single, 20-minute laboratory procedure. They question whether this assessment procedure is ecologically valid and also whether it can yield an adequate operationalization for such a central explanatory construct as security of attachment. Finally, some of the commentators question whether we can reasonably assume that secure attachment means the same thing at 12 months as at 18 months, an age at which many of the longitudinal studies have reassessed security of attachment with the same procedures.

When Goldsmith and I (Connell & Goldsmith 1982) coined the term "hour glass designs" to describe studies in which the ABC classification is the sole focus of hypotheses regarding antecedents and consequences of security of attachment, we were summarizing concerns similar to those of the contributors to the target article and the commentaries. These concerns regarding the ABC classification system and the developmental and cross-situational generalizability of the Strange Situation assessment procedure led us to call for an alternative assessment procedure and to suggest an alternative conceptual and data-analytic strategy. We believed such a strategy was necessary to achieve greater inferential clarity regarding key theoretical issues in the field, such as those raised in the target article and the commentaries.

We have not yet ventured outside structured laboratory paradigms for assessments of the attachment system; however, my colleagues and I (e.g., Connell, Thompson & Tero 1984) have used the factor analytic and structure modeling techniques that Goldsmith and I suggested and have derived a replicable, multidimensional representation of Ainsworth's interactive ratings in the "Strange Situation." In order to derive these dimensions (primarily from the separation and reunion episodes), we reanalyzed data from approximately a dozen Strange Situation studies (Connell & Tero 1983) including many of those reported in the Ainsworth, Blehar, Waters, and Wall (1978) book and those provided by the authors of the target article (e.g., Thompson, Lamb & Estes 1982). We are currently conducting a "mega-analysis" by combining data from these and other paradigmatically compatible studies of attachment over the second year of life. [Cf. Prioleau et al.: "An Analysis of Psychotherapy Versus Placebo Studies," *BBS* 6(2) 1983.]

The continuous dimensions emerging from our analyses have been employed to address a subset of the theoretical issues raised in the target article and the commentaries. Our theoretical perspective on the attachment system, not unlike the "traditional" perspective, emphasizes the role of the infant's emotions in "calling up" and regulating the social interactive behaviors observed in the Strange Situation. However, rather than assuming that emotion is a conceptually and operationally inextricable part of the attachment system, we have obtained independent ratings of emotional expression and then related individual differences in those emotional expressions to the social interactive behaviors of the infants toward the caretaker (Connell, Thompson & Tero 1984; Gaensbauer, Connell & Schultz 1983). This has allowed us to begin to separate component processes of the attachment system – in this case, emotions from social interactive behaviors – which were previously concatenated under the rubric of security of attachment. Our findings highlight the crucial and perhaps causal role played by the infants' emotions in the patterning of social interactive behaviors toward mother.

Using this multidimensional approach to characterize individual differences in components of the attachment system we have also begun to address another theoretical concern raised by certain commentators; namely, that normative developmental change may occur over the second year of life both in the meaning of security of attachment and, more generally, in the interpersonal processes which most centrally define the caretaker–infant relationship. In employing the ABC classification

system to assess individual differences in security of attachment in older infants, two implicit assumptions are being made: first, that the defining features of an adaptively functioning attachment system are invariant over the second year of life; and, second, that this system can be validly assessed in the same way near the end of the second year as at the beginning. We have examined these developmental issues in two different longitudinal data sets that included repeated Strange Situation assessments over the second year of life and independent ratings of emotion and social interactive behaviors (Frodi, Bridges & Grolnick 1985; Thompson, Lamb & Estes 1982).

In one study (Bridges, Grolnick, Frodi & Connell 1984), we examined relationships between independently rated dimensions of social interactive behaviors obtained from the Strange Situation procedure and the same infants' emotional responses in a structured series of toy play tasks with the experimenter. At 12 months, we obtained a significant positive relationship between this emotion dimension (coded for positive to negative emotions) and the social interactive dimension marked positively by "positive distance interaction" and negatively by "resistance toward mother." Thus, infants at 12 months, who were rated as emotionally positive in the toy play situations, tended to interact positively with their mothers from a distance and not resist contact with them during the Strange Situation assessment. The 12-month emotion ratings also predicted resistance in the Strange Situation at 20 months, but in the *opposite* direction. The infants with positively rated emotions at 12 months were *more* resistant toward their mothers at 20 months.

In the second study (Connell, Thompson & Tero 1984), we examined the stabilities of social interactive and emotional components of the attachment system, and a similar finding emerged. Of the group of infants who remained emotionally positive from 12½ to 19½ months (i.e., those who stayed above the median on an emotional dimension coded positive to negative), 62% shifted from below the median on resistance at time 1 to above the median on resistance at time 2. Thus, a significant proportion of infants who showed relatively positive emotional responses in the Strange Situation over the second year of life showed less resistance toward mother than their peers as 12½ months and more resistance at 19½ months.

These preliminary findings from our dimensional analyses of two components of the attachment system indicate that "resistance toward the mother," which is a salient marker of insecure or anxious attachment in the Ainsworth classification system and which appropriately signifies less adaptive functioning at 12 months, may actually signify adaptive functioning in the latter half of the second year of life. These findings led us to conclude that the two previously stated assumptions underlying the use of the ABC classifications are, at best, tenuous when applied to the older infants, and that alternative theoretical views of the development of the attachment system are needed. We are currently exploring the usefulness of the developmental theory of Mahler, Pine, and Bergman (1975) in this regard. These data and others emerging from our analyses fit with these theorists' view that over the second year of life the normal infant moves from a phase in which the mother is being used as a secure base for exploration in a relatively conflict-free way (one of Ainsworth's markers of secure attachment) to a phase late in the second year of life in which the infant is struggling with his growing understanding of both his mother's separateness and his own individuality. This struggle, termed "rapprochement," is, in their view, an important precursor of mature interpersonal relationships and manifests itself in bouts of resistance and ambivalent behavior toward the mother during this period. At this point, it would appear that the description given by Mahler et al. of developmental change in the mother–child relationship over the first 8 months of the second year of life is more compatible with the data than implicit assumptions of invariance in both the meaning and markers of secure infantile attachment over this period.

The findings of these initial studies indicate that our approach to the study of attachment is more than a mere flip of the Necker cube from a typological to a dimensional view of the same construct. Our approach is less bound by what many of the authors and commentators see as a tortuous set of theoretical assumptions regarding the evolutionary/biological bases of secure attachment. Our methodological approach is free from such constraints; hence the study of the attachment system becomes more open to empirical tests of competing theoretical views. Factors such as the quality and consistency of the caretaking environment and infant temperamental characteristics (Connell & Thompson 1986), rather than being advocated solely as contributors to *security* of attachment, can be treated as component processes of the attachment system along with infant emotions and social interactive behaviors. With a multidimensional characterization of each component process, the "fits" of alternative models which include longitudinal and contemporaneous relationships among the component processes may be estimated using structural modeling techniques. Finally, this dimensional approach to the study of component processes of the attachment system should yield a richer idiographic and normative portrayal of the attachment system as it functions in contexts other than the Strange Situation, in clinical populations, and in different cultures.

In conclusion, if we accept the Lamb et al. view that the empirical basis for the theoretical claims made by Ainsworth and her colleagues is weak, we are left with the question of whether the theory or the method has gone awry. It is clear that in nearly 20 years of theoretical and empirical work on this highly charged and operationally elusive construct, considerable progress as well as some mistakes have been made in both arenas. I and others, including Lamb et al. and most of the commentators, have suggested that alternative methodological approaches and theoretical views should continue to be encouraged and developed and should not be cast aside prematurely in favor of a single, predominant approach. Finally, I am encouraged that some consensus is being reached on a set of substantive questions to which various methods and theories can now be applied by their proponents, possibly in collaboration. The development and functioning of the attachment system in at-risk populations and in different cultures would seem to be two issues at the forefront of this emerging consensus.

Attachment as a motivational construct: I've seen these patterns before . . .

Martin E. Ford

School of Education, Stanford University, Stanford, Calif. 94305

Although Lamb et al. (1984) identify a number of limitations in the theoretical and empirical literature on infant attachment, there appears to be considerable agreement among attachment researchers on two major points: There are meaningful differences in infants' behavior in Ainsworth's (Ainsworth, Blehar, Waters & Wall, 1978) Strange Situation, and valid evidence links these individual differences to phenotypically distinct outcomes in later childhood. However, whereas Lamb et al. emphasize the plasticity of the individual differences captured in Ainsworth's ABC classification scheme and the importance of contextual factors in producing observed developmental continuities and discontinuities, researchers on the other side of the debate (e.g., Ainsworth et al. 1978; Sroufe 1983) focus on the stability of these individual differences and the contribution of the child's enduring characteristics to later developmental outcomes.

The primary purpose of this commentary is to show how a motivational conceptualization of attachment focusing on the child's perceptions of personal agency may be able to resolve these theoretical differences. In particular, I will argue that this

view helps one recognize the importance of both contextual and child characteristics in producing developmental continuity and change. It also specifies somewhat more precisely than present formulations the nature of the contextual and psychological processes that contribute to a child's attachment status and later manifestations of competence to which the attachment variable has been linked.

There is a growing literature on the importance of people's perceptions of personal agency in regulating behavior and promoting competence development. Although most of the current work on this key aspect of human functioning involves older children and adults (e.g., Bandura 1982; Deci 1980; Garber & Seligman 1980; Harter 1982; Weiner 1979), it is important to note that the article often cited as the main catalyst for this stream of theory and research deals primarily with infants' perceptions of control and competence (i.e., White 1959). Thus, the motivational processes that appear to form the basis for the infant's attachment relationships are processes that have been implicated in behavioral organization and development throughout the life span.

M. Ford (1985) has identified three major recurring patterns in the theoretical and empirical literature on perceptions of personal agency and has documented some of the extensive evidence linking each of these patterns to indices of psychological and social competence. Pattern 1 is characterized by the perception that one is an effective, controlling agent. In this motivational pattern the environment is perceived as responsive and the self is perceived as competent. This pattern is associated with feelings of confidence, enthusiasm, and hope. Pattern 2 is characterized by the perception that one is a powerless but potentially effective actor. In this pattern the environment is perceived as unresponsive to the self, but perceptions of personal efficacy are generally maintained through a variety of coping strategies (e.g., protesting, leaving the environment). Finally, Pattern 3 is characterized by the perception that one is both a powerless and ineffective agent. This motivational pattern is associated with behavioral uncertainty or passivity and with feelings of anxiety and depression. (Note that the fourth "cell" in this model – high perceptions of control and low perceptions of competence – is not treated as a separate pattern because it is apparently a transitional pattern; that is, one cannot easily maintain perceptions of personal control in the face of repeated failures for which one feels personally responsible.)

The basic premise of this commentary is that the three motivational patterns described above correspond at a fundamental level with the B, A, and C attachment classifications in Ainsworth's scheme, respectively. The correspondence is somewhat rough since the attachment categories were derived empirically from a relatively small sample, but there does appear to be sufficient evidence to justify the assertion that the three core patterns observed in the Strange Situation are manifestations of the three core motivational patterns described above. This evidence is summarized below.

In M. Ford's (1985) three motivational patterns, the basic contextual processes are contingent responsiveness and manageable challenge, and the key psychological processes are perceived control and perceived competence. However, because the meaning of the context is derived in large part from the application of an individual's self-referent perceptual filters, it is the latter set of processes which are decisive in behavioral regulation and competence development. This principle is quite evident in the attachment literature. Although caretaker sensitivity and responsivity appear to be important antecedents of a secure attachment relationship (e.g., Ainsworth et al. 1978; Sroufe & Waters 1982), the data are not as strong as one might expect from a model that does not place primary emphasis on the mediating psychological processes of perceived competence and control. Indeed, the more impressive data come from the predictive validity studies of Sroufe and his colleagues, which focus on motivationally relevant qualities of the *child* at different ages. In these studies the three motivational patterns are clearly revealed in the children's cognitive, emotional, and behavioral characteristics. Children earlier classified as B-group infants tend to show more enthusiasm and positive affect, engage in more symbolic and social play, and display more curiosity (Arend, Gove & Sroufe 1979; Matas, Arend & Sroufe 1978; Sroufe 1983; Waters, Wippman & Sroufe 1979). Importantly, they also score higher on measures of self-esteem, personal agency, self-regulation, and social competence (Londerville & Main 1981; Pastor 1981; Sroufe 1983). Alternatively, children classified as A-group infants tend to avoid their caregivers and later appear to be more hostile and socially isolated, whereas children classified as C-group infants tend to be indecisive in their interactions with their caregivers and are later rated as impulsive and helpless by their teachers (Sroufe 1983).

The motivational view of attachment also helps account for the body of research showing that the caregiving environments and developmental sequelae of the two insecurely attached groups are sometimes less differentiated than one might expect given the distinctive "look" of the A- and C-group infants in the Strange Situation. According to the personal agency perspective, both sets of infants perceive the environment as being relatively unresponsive. This implies that one should not expect large or consistent differences in their caregiving environments. Thus, *the distinction between A- and C-group infants primarily reflects different strategies of coping with stress rather than differences in the stressors themselves.*

Another reason one should not necessarily expect sharp differences between A- and C-group children (or some B-group children, for that matter) is that *the three motivational patterns are not mutually exclusive;* individuals may simultaneously manifest more than one pattern or alternate between patterns as the context changes. Thus, the real question for attachment researchers is not, Which kind of child is this? but rather, How prominent are each of the three patterns in this child? This suggests that researchers might be able to increase the validity of their Strange Situation assessments by scoring children in terms of the "amount" of each pattern they show rather than trying to fit every child into a single category.

This way of thinking may also account for much of the variation within the three basic attachment patterns. For example, A1 and C2 infants may be "purer" exemplars of motivational Patterns 2 and 3 than either A2 or C1 infants, who may show a mixture of all three patterns during the various episodes of the Strange Situation procedure. Similarly, B3 and B4 infants, who are sometimes distinguished from B1 and B2 infants for analytic purposes, appear to manifest a combination of Pattern 1 and Pattern 3 functioning (e.g., Easterbrooks & Lamb 1979).

Another issue that can be clarified using a motivational interpretation of attachment is the question of whether developmental continuities should be attributed to the context or to the child (Lewis, Feiring, McGuffog & Jaskir 1984). Because perceptions of control and competence are based in part on actual environmental conditions, one can generally expect significant contextual changes to be reflected in at least some competence-related assessments. However, this plasticity is balanced by the increasing stability that is likely to result as the child builds a coherent foundation of self-referent beliefs. For example, a child with a long history of Pattern 1 functioning may become relatively resistant to the potentially debilitating effects of stressful environments (Garmezy & Tellegen 1984). Similarly, children whose caregiving environments have been highly and consistently unresponsive may continue to show behavioral or emotional problems even when things improve for a time. Thus, although belief systems can change, even suddenly and dramatically, they tend to become rather resistant to change because of selective attention, confirmatory biases, and other social-cognitive phenomena (Nisbett & Ross 1980). Of course, the development of these belief systems depends on the differentiation of

self from others and the acquisition of rudimentary self-knowledge, but since these developments are well under way by the second half-year of life and quite advanced by the age of two (Lewis & Brooks-Gunn 1979), one can expect the three motivational patterns to be manifested quite early and to increase in coherence over time.

One should avoid assuming, however, that even strongly held beliefs about personal agency will be applied to every context. Although general perceptions of control and competence may be very potent regulators of a wide range of behaviors, people typically come to view themselves as having more personal agency in some contexts than others. Thus, it is quite possible for children to have different attachment classifications with different parents or other caretakers. Indeed, one can usually expect relationship-specific or task-specific perceptions of control and competence to be better predictors of behavioral effectiveness than general perceptions of personal agency (M. Ford 1985).

Finally, there is the question of the adaptiveness of the A, B, and C patterns of attachment behavior. Although the motivational interpretation presented here is not a biologically based theory, it does have clear implications for an evolutionary view of attachment behavior (D. Ford, in press; see also White's 1959 seminal paper on the biological origins of competence motivation). The motivational perspective asserts that *all three patterns of functioning are adaptive in contexts in which the reality is congruent with the perceptions involved in that pattern.* For example, avoidance may be a highly adaptive strategy in contexts that are generally unresponsive or distressing. Similarly, a mixutre of approach and avoidance may be quite adaptive in a caregiving environment that provides social-emotional or material resources in a somewhat inconsistent or unpredictable fashion. Thus, each individual needs to be able to call upon all three motivational patterns (i.e., all three patterns are species-appropriate). This does not mean that all three patterns will contribute equally to *longer-term* adaptations, however. Because the motivational consequences of each pattern are quite different, Pattern 1 functioning is generally more conducive to competence development than Pattern 2 functioning, which in turn tends to be somewhat superior to Pattern 3 functioning (M. Ford 1985; Lewis et al. 1984).

In sum, the motivational theory of attachment presented here appears to be able to resolve most of the controversial issues raised by Lamb et al. in ways that are mostly consistent with their observations and yet generally supportive of the Ainsworth and Sroufe et al. arguments regarding the importance of early development. Thus, if valid, this theory would bring us one step closer to understanding both the continuity of competence development and its openness to change (Ford & Thompson 1985).

The cross-cultural validity of the strange situation from a Vygotskian perspective

Marinus H. van IJzendoorn

Department of Education, Division of Theory and Methods, University of Leiden, 2300 RA Leiden, The Netherlands

Cross-cultural differences in distributions of children among attachment categories have raised the question whether the ABC classification represents alternative but equivalent pathways to maturity (Lamb, Thompson, Gardner, Charnov & Estes 1984). The American distributions led to the supposition that the modal B-category represented the optimal or normative pathway, the A- and C-category being considered the deviant patterns (Ainsworth, Blehar, Waters & Wall 1978). Research with the Strange Situation procedure in other countries has shown strongly different distributions (see Table 1). Lamb et al. (1984) therefore suggested interpreting these differences in a relativistic, nonnormative way.

In cross-cultural psychological research, three different broad trends or currents can be discerned (Cole & Scribner 1974; Van IJzendoorn & Van der Veer 1984). First, there exists a *relativistic* approach in which it is presupposed that different cultures lead to large differences in psychological adaptation (Lévy-Bruhl 1966). Different cultures provide their younger generations with different adaptive tools, including symbolic systems such as language. These generations therefore develop altogether differently. In attachment research this approach would lead to great doubt about the universal applicability of the concept of "attachment" and of the Strange Situation as its measurement procedure. Each culture has its own mode of caregiver–child interaction, and a uniform description of the relationship between caregiver and child through a uniform procedure would inevitably fail to do justice to specific culture-bound idiosyncrasies.

Second, there is a *universalist* current (Lévi-Strauss 1966) in which cross-cultural differences are not denied but are regarded as an unimportant veneer covering substantial similarities. Different uses of language, for example, can be traced back to the same underlying structure or competence. In attachment research this universalist approach leads to the claim that attachment develops in the same way and with the same structural characteristics (e.g., separation anxiety), despite notable differences in superficial aspects (e.g., nature and amount of physical contact). In fact, Bowlby (1969) and Ainsworth et al. (1978) take this point of view by situating the origin of a genetic "bias" for attachment in the more or less universal environment of evolutionary adaptation, and by regarding the crystallization period of different cultures as too short a time to lead to substantial differences in genetic equipment.

Third, there is the *cultural-historical* approach, inspired by Vygotskij (1962) and Lurija (1979). This Soviet-Russian approach, which strongly influenced such American psychologists as Bronfenbrenner (1979), Bruner (1983), Cole & Scribner (1974), and Wertsch (1985), tries to synthesize the relativistic and universalist points of view. The distinction between the lower and higher psychological processes is fundamental to this approach. The lower psychological processes, such as eidetic memory, are seen to belong to the basic evolutionary equipment of every human being. The higher psychological processes such as logical reasoning, however, develop quite late in phylogenesis. Mediation through language or other sign systems is characteristic for these higher processes. Because different cultures provide their members with largely differing sign systems, large cross-cultural differences in higher psychological processes can be expected. From a cultural–historical perspective the universal characteristics of the lower psychological processes are combined with great cultural variation of the higher processes.

The question, then, is whether attachment should be regarded as the outcome of lower or higher psychological processes. Because Vygotskij himself related the distinction between higher and lower processes largely to the development of communication through language, in his view attachment would be the product of lower psychological processes and would therefore have a universal nature. Elsewhere, we have criticized Vygotskij for his neglect of other sign systems than language; we have referred to the subtle nonverbal sign system through which caregivers and babies communicate (Van der Veer & Van IJzendoorn 1985). Against this background it seems unjustified to consider attachment as the expression of lower psychological processes. We feel justified in assuming that attachment behavior such as proximity-seeking, crying, and contact-maintaining is indeed the outcome of universally "programmed" lower psychological processes. Patterns of attachment, however, are the expression of nonverbally mediated higher processes that can be strongly culture-bound. The same

Table 1 (Van IJzendoorn). *Distributions of attachment categories in different cultures (percentages)*

Country:	United States[a]	Netherlands[b]	Israel[c] (kibbutzim)	Israel[d] (town)	Germany[e] (Regensburg)	Germany[f] (Bielefeld)	Germany[g] (Berlin)	Japan[h]	Sweden[i]
Attachment									
avoidant (A)	21	24	8	3	28	52	21	0	22
secure (B)	66	72	56	81	67	35	74	66	74
resistant (C)	13	4	33	16	5	13	5	34	4
Total n	106	136	82	36	38	46	?	29	51

Notes: [a]Ainsworth et al. 1978; 1-year-old American children with mother [b]Van IJzendoorn et al. 1984; Goosens, in preparation; 21-month-old Dutch children with mother [c]Sagi et al. 1985; 1-year-old Israeli children with mother (kibbutzim) [d]Sagi et al. 1985; 1-year-old Israeli children with mother (town) [e]Escher-Graub & Grossmann 1983; 1-year-old German children with mother (Regensburg) [f]Grossmann et al. 1981; 1-year-old German children with mother (Bielefeld) [g]Beller 1984; 1-year-old German children with mother (West Berlin) [h]Miyake 1984; 1-year-old Japanese children with mother [i]Lamb et al. 1982; 1-year-old Swedish children with mother

patterns of attachment could in different cultures fulfill different functions in the transaction between the child and its environment, but the lower processes that underlie these patterns are part of the genetically determined "equipment" of every human being.

From this Vygotskian perspective it is therefore admissible to search for universal attachment behavior, and its development in different cultures in the first years of a child's life, with a uniform measurement procedure such as the Strange Situation. "Deviant" patterns of attachment, however, have to be interpreted against the background of cultural differences. Every culture will have its own functional (and perhaps modal) pattern, which may deviate more or less from the American "standard." We hypothesize, for example, that in the Netherlands the modal B1-cateogry is more adaptive than its American counterpart, the B3-category. In the Netherlands – as in Germany (see Grossmann & Escher-Graub 1984) – much stress is laid upon the child's learning to fend for itself, and parents are afraid of spoiling the child at an early age with too much attention and explicit signs of love. In short, from a Vygotskian perspective one has to look not only for heterotypical continuity of adaptation across age and situation (Sroufe 1979), but across cultures as well. The same functional optimum could then be shown to be reached along different lines or patterns of attachment. Provided the classification is not normatively interpreted a priori, the Strange Situation procedure could play an important role in cross-cultural research as it taps the universal dimension of attachment behavior as an outcome of lower psychological processes.

Authors' Response

Convergent approaches to understanding strange situation behavior

Michael E. Lamb[a], Ross A. Thompson[b], William P. Gardner[c], and Eric L. Charnov[d]

Departments of Psychology, Psychiatry, and Pediatrics, University of Utah, Salt Lake City, Utah 84112[a]; Department of Psychology, University of Nebraska, Lincoln, Nebr., 68588[b]; Department of Psychology, University of Virginia, Charlottesville, Va. 22901[c]; Departments of Biology, Anthropology, and Psychology, University of Utah, Salt Lake City, Utah 84112[d]

The primary goal of our target article (1984) was to stimulate theorizing and research concerning the formative importance of parent–child relationships by reviewing the available evidence concerning the origins and developmental significance of Strange Situation behavior. These three commentaries, like many of those published with the target article, demonstrate the heuristic success of our article. All three commentators offer new and potentially fruitful suggestions regarding research and theory in this area.

Connell's commentary is exciting because it describes an independent but promising approach to the reanalysis and reinterpretation of Strange Situation behavior. Connell's component process approach, as described here and more fully in Connell (1985), underscores the crucial role played by emotional arousal in shaping the pattern of social interactive behavior in the Strange Situation. His argument is buttressed by two recent reports: one by Thompson and Lamb (1984) showing that individual differences in emotionality explained much of the variance between Strange Situation groups and another by Thompson and Connell (1985) demonstrating that in two short-term longitudinal studies, emotion was a better predictor of subsequent social-interactive behavior than the reverse. Clearly, further research on the functions of affect in mother–infant interaction patterns is called for, and the analytic strategy pioneered by Connell merits additional exploration in this regard. The same approach may also be helpful for pursuing work in social-emotional development in the second year of life, as suggested by Connell. Toddlerhood has been largely ignored by developmental scientists, despite the manifest importance of this transition from infancy to the preschool period for those interested in ontogenetic continuities and discontinuities.

We are also impressed by the substantial convergence between the ideas raised by **Van IJzendoorn** and the arguments we have raised regarding the interpretation of cross-cultural differences in Strange Situation behavior. Our arguments were not included in the target article, where space was at a premium, but are discussed at some length in our recent book (Lamb, Thompson, Gardner & Charnov 1985). There are evident differences between

our approach and Van IJzendoorn's but we agree on the need to recognize that experiential (and perhaps constitutional) differences between cultural groups may affect Strange Situation behavior *at least as much* as variations in maternal sensitivity may do so. Thus the high levels of avoidance among German babies (Grossmann, Grossmann, Huber & Wartner 1981) may reflect culture-specific beliefs in the importance of early independence training rather than culture-specific deficits in maternal sensitivity, whereas the high levels of distress and resistant behavior among Japanese and Israeli infants (Miyake, Chen & Campos 1985; Sagi, Lamb, Lewkowicz, Shoham, Dvir & Estes 1985) may reflect unfamiliarity on the part of infants with brief mother–child separations (Japan) or strangers (Israeli kibbutzim) due to distinctive patterns of childrearing. Such possibilities underscore the need to understand the psychological meaning of the Strange Situation procedure to infants tested in it.

This issue is no less important in studies of within-culture differences in Strange Situation behavior, as **Ford** implies in his commentary. Unfortunately, however, Ford's speculations are weakened by his uncritical reliance on the extant literature. In the target article, we showed that the relationship between parental behavior and infantile Strange Situation behavior was much weaker than had previously been claimed. Only a general relationship between socially desirable patterns of maternal behavior and "secure" Strange Situation behavior has been demonstrated in the U.S. Furthermore, it is not clear whether prior rather than contemporary patterns of parent–child interaction are significant, because researchers have failed to investigate this question systematically. Additional research published since preparation of the target article confirms our initial conclusion, and is discussed at length in our recent book (Lamb et al. 1985). In light of this, **Ford**'s commentary is perplexing, because he takes for granted the existence of specific associations between parental and infantile behavior that have yet to be demonstrated. Likewise, he overstates the available evidence concerning the predictive validity of Strange Situation behavior which (as we showed in the target article) is still quite weak, especially when there is instability in family and childcare circumstances. In addition, his personal agency perspective, which in many respects echoes our earlier interpretation of Strange Situation patterns in terms of social expectations (Lamb 1981a; 1981b), becomes frustratingly vague by virtue of comments such as "individuals may simultaneously manifest more than one pattern or alternate between patterns as the context changes" and "[some classification subgroups] may show a mixture of all three patterns during the various episodes of the Strange Situation." Presumably, these statements are designed to acknowledge and take into account the weakness of the empirical data, but the net effect is to render the whole approach unclear. We believe that it is fruitful to explain Strange Situation behavior in terms of infantile cognitions and motivations, but these interpretations will not be helpful unless the speculations are both clear and consistent with the available evidence.

It is interesting that none of the commentaries we have elicited focus on the increasingly widespread claims about the relationship between infantile temperament and Strange Situation behavior (cf. Sroufe 1985). In the interests of brevity, a discussion of this issue was excluded from our target article. In our book (Lamb et al. 1985), however, we showed that there is even less support for a temperamental interpretation of Strange Situation behavior than there is for an interpretation in terms of prior patterns of parental behavior. Unfortunately, the failure of researchers to test realistically complex models of temperament–behavior relations mandates caution in interpretation of these data, at least for the present.

In sum, recognition of extant ambiguity concerning the origins, stability, and developmental significance of individual differences in Strange Situation behavior should serve to stimulate rather than terminate efforts to understand the formative significance of parental behavior, cultural rearing practices, and infantile temperament. Patterns of Strange Situation behavior clearly reflect individual differences in "security of attachment," but they also appear to reflect other aspects of infantile experience and perhaps temperamental variations as well. The goal should surely be to develop a less noisy measure of attachment security, in part by continuing work designed to understand individual differences in Strange Situation behavior. These three commentaries, by suggesting fruitful new approaches to research on these issues, demonstrate the success of our efforts to stimulate precisely such exercises.

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Commentary on Derek Bickerton (1984) The language bioprogram hypothesis. BBS 7:173–221.

Abstract of the original article: It is hypothesized that creole languages are largely invented by children and show fundamental similarities, which derive from a biological program for language. The structures of Hawaiian Pidgin and Hawaiian Creole are contrasted, and evidence is provided to show that the latter derived from the former in a single generation. A realistic model of the processes of creole formation shows how several specific historical and demographic factors interacted to restrict, in varying degrees, the access of pidgin speakers to the dominant language, and hence the nature of input to the children of those speakers. It

is shown that the resulting similarities of creole languages derive from a single grammar with a restricted list of categories and operations. However, grammars of individual creoles will differ from this grammar to a varying extent: The degree of difference will correlate very closely with the quantity of dominant-language input, which in turn is controlled by extralinguistic factors. Alternative explanations of the above phenomena are surveyed, in particular, substratum theory and monogenesis: Both are found inadequate to account for the facts. Primary acquisition is examined in light of the general hypothesis, and it is suggested that the bioprogram provides a skeletal model of language which the child can then readily convert into the target language. Cases of systematic error and precocious learning provide indirect support for the hypothesis. Some conjectures are made concerning the evolutionary origins of the bioprogram and what study of creoles and related topics might reveal about language origins.

Grammar growth and parameter setting: Computation and creoles

Robert C. Berwick

MIT Artificial Intelligence Laboratory, Cambridge, Mass. 02139

Derek Bickerton (1984) urges us to consider "a parameter-setting model" as the right way to forge a unity between his model of creole language formation and the Chomskyan viewpoint. As he notes, it is "only a preliminary sketch" (p. 217). Fortunately, more than a preliminary sketch of just such a parameter-setting model is already available – the explicit computer model set out in Berwick (1982). The results of that study show that there is considerable complexity to a model of "parameter-setting" – far more than Bickerton's informal comments would indicate. To summarize the results of that work: First, one can build an explicit parameter-setting model that learns grammars from positive examples. Second, formal study of parameter setting has revealed several general principles of parameter learning; several researchers are already using this emerging theory of parameter setting to guide their work on actual child language acquisition. Third, this same model has shown just how tricky it is to invoke developmental ordering as a substitute for innate principles (as Bickerton 1984r, does in his reply to Hornstein 1984). Finally, the work shows, contrary to Bickerton's assertion, that systematic errors (of the sort actually observed) are compatible with a parameter-setting model; indeed, the model explains certain observed errors that children actually make.

Modeling acquisition. It's not enough, as Lightfoot (1984) and other commentators remarked, just to say that "processes of modification" suffice to guide the creole grammar to other natural grammars – one has to show that this is so. Berwick (1982) is a step in this direction. The key proposal is a specific model of language acquisition based on trying to parse sentences of a language and constructing new rules when old rules fail. This computer model actually works. It can learn from simple positive example sentences a majority of the hand-coded rules designed for the Marcus parser (Marcus 1980). Along with this computer-based account, Berwick advances a formal framework in which to analyze "parameter-setting" models. This uses a complexity metric based on the number of decisions required to fix a rule system. Berwick shows that this developmental model actually explains the observed distribution of natural segmental (sound) systems – why we see only some clusters of consonants and vowels in natural languages and not others. This work also shows how the same metric can be used to explain the development of nonterminal categories like NP, VP, PP, and so forth within the $\bar{X}$ system of phrase structure.

Parameters and acquisition order. There's more to parameter setting than meets the eye. Bickerton's parameter model isn't novel. It has already been the subject of much research and debate. (A recent conference in the linguistics department at the Univeristy of Massachusetts at Amherst, May 1984 [Williams & Roeper 1984], was devoted largely to this topic.) Again, formal analysis has proved helpful here. It's not enough just to set parameters in any order, given some sequence of evidence. Given positive evidence, parameters must be set in a particular order – an order such that it is always possible to get positive evidence to refute a possibly incorrect guess. One must always ensure that a new parameter setting never allows for a possible "narrower" parameter setting. This so-called Subset Principle discussed at length in Berwick (1982) and by Williams (1981) imposes an ordering on parameter setting that actually accounts for part of the observed pattern of phrase structure acquisition. Note that this does not presume that every individual child must do this work of ordering; this would amount to a subtle form of learnability buck-passing (see the next paragraph). Rather, the ordering is fixed over phylogenetic time. It would be interesting to extend this principle to the case of creoles. More recently, Wexler and Manzini (1984) have advanced refinements of the Subset Principle that give promising insights into actual patterns of child language acquisition. Parameter setting is here to stay.

Developmental sequences and input data. A good example of the salutary role of precise modeling comes up in Bickerton's (1984r) reply to Hornstein's (1984) point that different input data orders might lead to different grammars. Presumably, as several commentators noted, we want children exposed to different sequences of examples to attain the same final state (grammar). Bickerton says "because of developmental factors, a child at any given stage can only attend to a given subset of the data" (p. 216). But what exactly does this mean? Evidently, the "developmental factors" conspire so as to turn the child's attention to the right input sentences at just the right time. Note that this actually puts some of the burden of learning on whatever it is that pays attention to the input. To prevent this from becoming mere buck-passing, one has to show that in the examples that actually arise the "attentional" model can work without demanding too much intelligence on the part of the device that is doing the attentional filtering. Again, this is part of Berwick's (1982) demonstration: The acquisition device can only acquire one new grammar rule at a time. This limit imposes a plausible attentional filter on the input – enough to make acquisition stable relative to different orders of presentation of example sentences.

Parameter setting and systematic errors. Bickerton says that all the parameter-setting models he is familiar with have great difficulty in accounting for systematic errors that appear in acquisition. But this is not true of Berwick's (1982) model. Rather, this model gives a natural explanation for the appearance of such mistakes. Errors can arise if (correctable) mistakes are made in fixing the category identity of lexical items. These errors of feature identification interact with (correctly formulated) grammatical rules and a system of skeleton phrase structure rules called $\bar{X}$-rules to yield surface sentence errors. For example, consider Roeper's observations (1980) about the use of for in child constructions. Evidently for-NP constructions, e.g., This one is for you; There's one for Mom appear first in purposive constructions. Soon after, for-S constructions appear: I want for you hold it. If we say that for's feature assignment is learned from the for-NP case, namely, as a purposive, and if we assume (following the $\bar{X}$ theory and the model in Berwick 1982) that NP and S share some features, then the extension to S's follows. But then, any purposive clause will do, and so we find errors such as, The milk is for to drink. To give just one other example, the PRO drop parameter (the omission of Subjects in Romance languages like Italian and Spanish) has been used successfully to explain corresponding errors in child acquisition in English.

In sum, not only are parameter models compatible with facts about mistakes in acquisition, they may fruitfully explain them.

Creoles and computation. Plainly, one direction to go would be to apply the computer model to creole-type input and see what happens. Particularly intriguing is the problem of conflicting evidence. The currently implemented model has no way to handle this kind of situation. One solution is something like Jackendoff's (1983) idea of preference rule systems: a set of competing rules that alternate between multistable equilibrium points. Preference rules have been used to account for perceptual analyses of visual scenes (the Necker cube illusion) and alternate perception of musical rhythms; they may prove to be good models for creole-type situations where distinct grammars are in collision.

Bickerton's creole cooking: Where's the beef?

John J. McCarthy

Department of Linguistics, South College, University of Massachusetts, Amherst, Mass. 01003

Bickerton (1984) claims that creole languages enjoy a special place as exponents of Universal Grammar (UG), and in support of this he propounds two theses, one considerably weaker than the other. The stronger one, which I will address first, is that the properties of UG are exhausted in the properties of creoles. The weaker hypothesis – that creoles merely represent unmarked settings of parameters of UG – we will return to below.

Two of the first-round commentators, Marantz (1984) and Woolford (1984), observe that there are interesting, reasonably well understood, and apparently universal properties of languages that are represented poorly, if at all, in Bickerton's menu of creoles, and for this reason the stronger hypothesis cannot be maintained. Bickerton evidently feels that he sufficiently dismisses this criticism when he says that Chomsky said the same thing, but more cleverly. In view of this, it is not surprising that Bickerton has excluded large areas of linguistic structure from the bioprogram:

> Children . . . have a core grammar that is moderately specified but is obviously not specified down to lexical items, not specified down to phonological rules, not specified down even to low level syntactic rules, but which is highly specified with regard to the core items of syntax, the core items of semantics. (Bickerton 1979, p. 16)

On the contrary, I argue that, once we turn from such quasi-functional domains as the selection of aspectual over tense systems (scarcely a "core item") to the purely formal beef of linguistic theory, the commentators' original criticism is devastating to Bickerton's stronger claim. Let us consider in particular the most formal domain of all in linguistic theory – the study of word and sound structure under the rubrics of morphology, phonology, and phonetics.

It is a well-established fact (Fauquenoy 1974; Hancock 1980; Kay & Sankoff 1974; Whinnom 1971) that creoles in the stage immediately after their creation – the stage with which Bickerton is concerned – are extremely impoverished morphologically, morphophonemically, and phonologically. Where any of these domains has been elaborated, as in Haitian Creole, the elaboration has developed subsequent to creolization (Valdman 1977). Creoles characteristically lack any but the most rudimentary word formation processes, they have few if any phonological or morphophonemic alterations, and their sound systems lack many of the phonological contrasts that we find quite often in the languages of the world. Yet research on morphology and phonology in languages other than creoles provides abundant evidence (cf. Aronoff and Oehrle, eds., 1984, for a recent review) that these domains have a complex and well-ordered internal organization constrained by a rich universal theory that is little attested in creole languages.

Consider an extremely simple example along these lines. No creole has been reported to observe a phonologically relevant distinction between θ (the initial sound of thin) and s, a distinction that obviously operates in English and that is characterized by a difference in the value of the distinctive feature [strident]. If UG is coextensive with the intersection of creole grammars, as in the strong form of Bickerton's hypothesis, then it follows that the feature [strident] and the phonological distinction derived from it must be an extralinguistic imposition on the grammar of English rather than an option provided by UG. This is true if even one creole lacks the distinction. Apart from the inherent implausibility of this claim, it leaves unexplained the following facts: [strident] in English has a systematic cross-classificatory function (it also appears in the voice pair δ [the sound in that] and z); [strident] is essential to the statement of the rule (inter alia) for pronouncing the plural suffix (contrast paths and passes); and stridency contrasts are found in many other languages.

Similar examples and others of much greater complexity can be adduced almost endlessly from English or other languages, all pointing to the conclusion that large, impeccably supported domains of linguistic theory and therefore of UG are virtually unrecoverable from creole languages. It follows, then, that the strong form of Bickerton's hypothesis, by which the grammars of creoles exhaust UG, is simply false, evidence from doubtful events in the early history of Hawaii notwithstanding. What of its weaker form, however? Bickerton suggests that, if creoles don't exhaust UG, perhaps they represent the unmarked settings of the parameters. Although this sounds interesting, in practice it is difficult to distinguish it from the entirely vacuous claim that creoles contain properties determined by UG – something that is true by the definition of UG and the fact that creoles are natural languages. Bickerton's own example – the use of aspectual distinctions in creoles – is a perfect illustration of this trap of vacuity, because no parameters embedded in an articulated and well-supported theory are or have been proposed by which some languages distinguish tense and others aspect.

Granting that one could nevertheless tease out all the parameters, what are we left with? Well, in the case of the feature [strident] the markedness account has some plausibility: Since Jakobson's early work it has been known that a phonological distinction between θ and s contributes to the markedness of a language, and it has also been known that the loss of marked phonological oppositions is a characteristic of language simplification with much broader scope than creoles alone. But if we look instead at the morphological and morphophonemic domains, where creoles are essentially without representation, then we are forced to what must surely be an absurd conclusion. If creoles invariably exercise the unmarked options of UG, as Bickerton maintains, then it must be unmarked for languages to lack morphology and morphophonemics entirely. This means that most of the world's languages are in the perilous state of having highly marked grammars and, far more seriously, that the whole theory of morphology and morphophonemics must be a theory of how languages articulate a marked option, with the unmarked option reduced to the single abjuration "Don't!"

The conclusion is that the beef – the purely formal domains of linguistic theory – just isn't there in Bickerton's proposals, and that granting a special status to creoles is unsupportable.

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Author's Response

"Grammar growth" – what does it really mean?

Derek Bickerton

Department of Lingistics, University of Hawaii at Manoa, Honolulu, Hawaii 96822

McCarthy's title is singularly appropriate for his commentary. It was, of course, lifted from a slogan in a fast-food advertising war; just like the combatants in that war, McCarthy is too busy hustling his own brand of hamburger to bother about what really goes into brand X.

For instance, there is the claim that "selection of aspectual over tense systems" forms part of the typology of creoles (hence of the bioprogram), a claim looming so large in **McCarthy**'s imagination that he pours scorn on it twice. It's a shame to see good scorn wasted, but alas, the claim is one I never made. Quite the contrary – I've been pointing out at least since 1974 that although this claim is among the hardiest perennials in creole studies, it is really quite incorrect, because creole tense-mood-aspect systems are built out of one marker of tense, one marker of mood, and one marker of aspect. It is necessary to get one's facts straight if one's criticisms are to be taken seriously.

Which brings us to **McCarthy**'s scholarship, perhaps best illustrated by the claim, buttressed by four citations, that "creoles in the stage immediately after their creation . . . are extremely impoverished." In fact, for most creoles there is no historical evidence whatsoever that would show us what they were like "immediately after creation," and what the works he cites say about this stage is either unsupported speculation (the case of Kay and Sankoff 1974) or else quite incompatible with McCarthy's claim; for example Fauquenoy (1974,p. 29) explicitly states that the earliest known text in Guyanais dates from 1848 (the language itself began at least two centuries earlier). In those rare cases where wo do have earlier texts (1718 for Sranan, 1740 for Papiamentu) the structure of these languages appears to have been no less complex then than it is today. As for the Haitian determiner system claimed by Valdman (1977) to represent a postcreole elaboration, this claim too is a speculative one, with no textual foundation, but based on some very questionable assumptions about the origins and dissemination of French creoles in general. Linguists in other fields would be indignant, and rightly so, if strangers to those fields were to claim expertise on the basis of a handful of ill-digested and ill-understood articles – but for some peculiar reason (linguistic snobbery, perhaps, dating from the days when creoles weren't "respectable" languages) it seems to be quite okay for anyone to throw normal scholarly caution to the winds and become Instant Creolists.

When you've scraped off its rhetorical sauce, the McCarthyburger proves to contain little beyond the notion that because creoles are "essentially without representation" in morphology and morphophonemics, they must be too impoverished to be representatives of U(niversal G(rammar). Well, I never made any claims about creole phonology, which renders **McCarthy**'s paragraph on θ and s quite irrelevant. Also, when McCarthy says "morphology" I assume he must mean *bound* morphology, or his remarks do not make sense. If having little or no bound morphology puts creoles in a minority of the world's languages (a minority that includes English and Chinese, by the way), what has this to do with markedness? Nothing, unless you measure markedness by counting heads, as in the tradition of Greenberg (1966). As I thought every generativist knew, it is only in terms of some theory of language that one can even talk about markedness and unmarkedness. Thus there is nothing in principle that would prevent creoles from instantiating unmarked options of UG *and* being the only languages that instantiated those options (I don't for a moment think this is the case, but it wouldn't affect my argument even if it were).

In contrast, **Berwick**'s commentary does raise substantive issues, but in a puzzling manner. It is indeed interesting to know that Berwick has already developed a model of parametric grammar resembling that which I briefly sketched in my Response (Bickerton 1984r) and that he has obtained some apparent confirmation of this model from acquisition studies. What is not clear to me, and doesn't seem particularly clear to Berwick either, is exactly how all of this relates to the central issue in the target article: What exactly do creole languages represent? Again, Berwick's closing remarks suggest that he has not grasped those sections of the target article that dealt with the empirical circumstances of initial creole formation. There is no reason to think that the problem of conflicting evidence is central to those circumstances and thus no grounds whatsoever for supposing that some "now you see it, now you don't" model of alternating preferences could account for creole grammars. Creole languages are not mixed languages, made up of bits of other people's grammars; they are sui generis, representative of a typology that is quite unmistakable once you are familiar with a few examples of it.

The central problem of creole formation is a poverty of the stimulus that went far beyond anything to be found in normal acquisition. One might suppose that the resulting deficit could have been made up from other input accessible to the child, but this does not seem to have happened, except perhaps in the case of some options left open in UG. In fact, no nativist should expect the deficit to be made up in this way. Whether the child's endowment consists of a single bioprogram or a parametrized grammar with unmarked options, it must be a whole lot easier for that child simply to build a new language with the aid of such an endowment than to learn a fully formed language or languages and then transfer this knowledge to the nascent creole.

Unless, of course, one assumes a model of parametric grammar in which all switches are initially in neutral positions and for which positive evidence of some kind is required before *any* setting can be fixed. **Berwick** isn't clear about whether or not his model is of this type, but some of his terminology, such as his reference to "a possibly incorrect guess" on the part of the child, suggests that it may be. As I see it, models of UG fall into one of the following three classes: (a) parametrized, switches at neutral, (b) parametrized, switches at unmarked settings, and (c) nonparametrized (a unique biological grammar). (There could, of course, be "hybrid" models, an interest-

ing possibility that I will not explore here; all parts of such models must, however, be drawn from the set of possibilities in (a)–(c).) My claim is that the facts of creole genesis automatically rule out (a). My initial claim was that (c) – my "stronger hypothesis" in **McCarthy**'s terminology – was the appropriate class. In my Response I conceded that (b) – my "weaker hypothesis" – might be preferable. I now think that the latter move may have been premature.

The case for (c), at least as a guiding assumption in research, has perhaps been most clearly stated by Borer (1983, p. 2):

> It is a desirable step forward to try and restrict the class of possible parameters. The strongest claim in this respect would be that there are *no* language-particular choices with respect to the realization of universal processes and principles. Rather, interlanguage variation would be restricted to the idiosyncratic properties of lexical items. These idiosyncracies, which are clearly learned, would then interact with general principles of UG in a particular way. This interaction would result in vastly different systems.

It follows inevitably from any such view that if a situation arises in which all that is passed on to children is a minimal lexicon purged of many, perhaps most, of its idiosyncratic properties, what will emerge will represent the processes and principles which characterize our biological program for syntax with less interference from idiosyncratic properties of languages than elsewhere. This is exactly what the target article claimed, at least with respect to the more radical creoles such as Saramaccan.

Berwick cites Wexler and Manzini (1984) in support of his claim that "parameter setting is here to stay." He may have been too hasty in this claim, and he certainly misinterprets Wexler's position as it is set forth in a paper delivered at the same conference (Borer & Wexler 1984). Berwick rejects the idea that "developmental ordering" plays a part in language acquisition (I never claimed this as a substitute for innate principles, contrary to what he states). Borer and Wexler, on the other hand, explicitly state that "certain principles mature. The principles are not available at certain stages of a child's development, and they are available at a later stage. . . . Like any other instance of biological maturation, the principles take time to develop."

On the Borer–Wexler view, it is simply incorrect to state, as Berwick does, that "the 'developmental factors' conspire so as to turn the child's attention to the right input sentences at just the right time." Children's attention does not need to be turned to anything at any time, at least insofar as the acquisition of syntax is concerned, because their syntax will just go on maturing as inevitably as their secondary sexual characteristics. To put it simply, at each level of syntactic maturation children will develop additional principles of UG which in turn will enable them to make sense out of more (and more complex) morphological and lexical properties; these, on being acquired, will interact with the syntax so as to yield the grammar peculiar to English (or whatever the target is). This view is fully compatible with my own position. At present it is still a minority view in the field of acquisition, but there are strong arguments in its favor.

The sad thing about both commentaries is that neither really tries to deal with the problem of how new languages come into existence. A frequent response by generativists to their critics is that it is pointless to pick holes in theories; the only appropriate response is to produce alternative theories that give a better account of the facts (see e.g., the remarks by Chomsky 1981, [p. 149], on Matthews 1980). The language bioprogram hypothesis will continue to stand until generativists begin to practice what they preach.

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Commentary on **Endel Tulving (1984) Précis of *Elements of episodic memory*. BBS 7:223–268.**

Abstract of the original article: *Elements of episodic memory* (Tulving 1983b) consists of three parts. Part I argues for the distinction between episodic and semantic memory as functionally separate albeit closely interacting systems. It begins with a review of the 1972 essay on the topic (Tulving 1972) and its shortcomings, presents a somewhat more complete characterization of the two forms of memory than the one that was possible in 1972, and proceeds to discuss empirical and theoretical reasons for a tentative acceptance of the functional distinction between the two systems and its possible extensions. Part II describes a framework for the study of episodic memory, dubbed General Abstract Processing System (GAPS). The basic unit in such study is an act of remembering. It begins with the witnessing of an event and ends with recollective experience of the event, with related memory performance, or both. The framework specifies a number of components (elements) of the act of remembering and their interrelations, classified under two broad categories of encoding and retrieval. Part III discusses experimental research under the label of "synergistic ecphory." Ecphory is one of the central elements of retrieval; "synergistic" refers to the joint influence that the stored episodic information and the cognitively present retrieval information exert on the construction of the product of ecphory, the so-called ecphoric information. The concept of encoding specificity and the phenomenon of recognition failure of recallable words figure prominently in Part III. The final chapter of the book describes a model, named the synergistic ecphory model of retrieval, that relates qualitative characteristics of recollective experience and quantitative measures of memory performance in recall and recognition to the conjunction of episodic-memory traces and semantic-memory retrieval cues.

Remembering experiences and the experience of remembering

Robert G. Crowder

Department of Psychology, Yale University, New Haven, Conn. 06520

Above all, one must not take for granted the title and opening sentences of Tulving's (1983) book. With the term "episodic" Tulving is not just announcing a book about memories planted in the laboratory, rather than about the study of prior world knowledge from the outside. That straightforward distinction, between episodic and semantic memory, respectively, is what he calls the "heuristic" distinction. The heuristic distinction is now so widely accepted, it is easy to get well into the book before realizing that Tulving proposes here a radical new focus for episodic memory. That memories should be both spatially and temporally articulate in the rememberer's personal experience is a necessary but not a sufficient condition for episodic memory, in the sense used here; there must also be "recollective experience." Episodic memory entails "the ability to experience again now, in a different situation and perhaps in a different form, happenings from the past, and know that the experience refers to an event that occurred in another time" (Tulving 1983, p. 1). This subjective *experience* of having a past event from one's life projected into the present is not just an additional criterion for defining episodic memory, it is the controlling definition of episodic memory, and so of the book's domain; "recollective experience should be the ultimate object of interest, the central aspect of remembering that is to be explained and understood" (Tulving 1983, p. 184). Tulving is quite right that this aspect of remembering has been neglected almost entirely by experimental psychologists, though apparently it has been a preoccupation of philosophers.

The heuristic and experiential senses of episodic memory are brought into sharp contrast when Tulving deals with the exciting new literature on unconscious indices of (what I now have to call) memory for laboratory events. An influence of prior exposure to a word can be measured by facilitation in various perceptual tasks, including lexical-decision priming, word identification thresholds, and spelling tests, all without the subject's knowing that his memory for the earlier presentation is being tested at all. These unobtrusive tests do not, however, *require* memory for the laboratory event in order to be performed, and therefore Tulving calls them semantic memory tasks. That is, one can decide that "foible" is an English word whether or not it has been encoded in the last hour; it's just that this decision will be measurably faster if it has occurred in the last hour than if it has not. Tulving's argument for separate semantic and episodic memory systems is largely based on dissociations between experimental effects for the same items when tested in these semantic-memory tasks or when tested in traditional episodic (recall or recognition) tasks. On the other hand, the *facilitation* on the semantic-memory tasks that results from a word's prior exposure presents something of a problem. Those of us comfortable with the heuristic distinction would have assigned this facilitation to episodic memory. Tulving (Chapter 6) seems unwilling to allow this much cross-pollution of the two systems. Instead, he proposes (p. 110) that the priming and facilitation effects in semantic tasks are due to gains in cognitive skills, in procedural, not semantic memory. Skills and all procedural memory were, early in the book, isolated from episodic memory by an even more basic division between procedural and declarative (episodic and semantic) information. Tulving's assignment of perceptual facilitation effects in semantic-memory tasks to perceptual skill rather than to episodic memory leaves him in an awkward, internally inconsistent position regarding a task such as lexical decision. On page 88 (Table 5.4), the lexical decision task, measured by reaction time, is taken to be a semantic-memory task. On page 34, however, the same reaction-time index of lexical decision is described as a measure of cognitive skill (with accuracy as the measure of declarative, semantic memory). The assignment of accuracy and speed measures of the same lexical decision task to different memory systems is eccentric, and ignores the fact that any accuracy measure can be pushed up and down by lenient or stringent response deadlines. The treatment of word-fragment completion (for example, on and around p. 104) is troubled in the same way. Thus, the dissociations displayed in the first section of Tulving's book may tell us more about the procedural/declarative distinction than about the semantic/episodic distinction.

So sharp a distinction between procedural and declarative memory (including both episodic and semantic) is highly debatable on other grounds. Tulving makes the capacity for drawing inferences central to the semantic memory system (p. 44), and thus explicitly not procedural. However, if one examined an explicit process model for making inferences, it is very likely to resemble a procedural skill. Teachers of logic would probably think so. From another perspective, there is increasing interest these days in the possibility that episodic-memory tasks are heavily infused with procedural memory. Paul Kolers's (1979) work on memory for sentences presented in unusual typographies is hard to interpret otherwise. Skill and knowledge may, after all, be complementary aspects of the same thing rather than abruptly dissociated mental systems.

How people remember to do things in a certain order is never addressed, as far as I could determine. On page 61, it is clear that immediate memory (digit span) is not included in the episodic system, but with no explanation as to why. Should we assume that other serial memory systems are excluded as well?

If so, perhaps it would be on the grounds that they, too, represent some kind of procedural learning (although that argument would run afoul of Tulving's other criteria for the procedural/declarative distinction). This apparent dismissal of the serial-order problem is typical of what faces the reader of this book, as he sees his favorite research literatures removed from what he used to think was episodic memory for various reasons – because the measure of performance is priming in a semantic-memory task, because the task is procedural, or perhaps only because the retrieval does not carry the subjective flavor of recollective experience.

Obviously, Part I of the book offered some real shocks and provocations for at least one reader. If it has done this for others, it is an immensely valuable contribution to the field, because we have been worrying too little about what it is we have been studying. And once past these highly controversial proposals, Parts II and III are delightful. With masterfully clear explanations everywhere, with charming and persuasive "thought experiments," with illuminating personal commentaries on the process of research in its widest social setting, these remaining eight chapters place a coherent theoretical and empirical context on Tulving's truly *indispensable* contribution to the science of human memory. Especially in Part III, where Tulving reviews his landmark experiments of the 1960's and 1970's, I had the recollective experience of spending time, once again, with old friends and celebrating how very well they had aged.

Systematizing cognitive psychology

Marcel Kinsbourne

Department of Behavioral Neurology, Shriver Center, Waltham, Mass. 02254

Tulving (1983) contends that multiple systems subserve memory and learning and he thinks he has discovered two. Before adjudicating in customary manner – what evidence, how else interpretable – let us understand the theoretical importance of the issue. Failing that, we risk submerging the work in fatuous generalities such as "heuristic," "promising," or some other lethal epithet.

Cognitive psychologists choose their issues by the "Why don't we. . . ?" method. Why don't we find out which causes what, whether this works like that, whether they do better than the others? It is the agony and also the ecstasy of graduate studies that there is no yardstick by which one can determine which line of inquiry deserves precedence over another. Tulving identifies this stage of our science as "pretheoretical." It is analogous to pre-Darwinian biology. Linnaeus described animals, but could not define species because he could not tell which differences made a difference. Psychologists can only group the performances that they measure into categories by conventional ("folk") verbal labels (e.g., "memory" and "learning"). Tulving urges us to attempt to systematize memory (and by implication, all cognition); to determine experimentally how one performance relates to another.

This is the enterprise: to organize cognitive psychology as Darwin organized zoology. For Darwin the organizing principle was evolutionary transition. For us it will be the differentiation of neural systems subserving cognition. This effort is so alien to traditional psychological thinking that it has gone unnoticed, because it is not understood. Certainly its rules of evidence cannot be derived unchanged from the existing experimental literature.

If performances (P1, P2) are controlled by separate systems, they must vary independently. It follows that:

1. Success in either does not predict success in the other (Tulving's stochastic independence).

2. P1 and P2 should have some different determinants (crossover interaction).

These outcomes are necessary to diagnose separate systems. Are they sufficient?

Consider the following Gedankenexperiment: A word list is studied. At retrieval, cueing condition C1 (e.g., semantic) is used. The subject then attempts further recall under cueing condition C2 (e.g., phonetic). Will the probability of retrieval under C2 differ from the base rate of C2 retrieval probability (established in a control condition)? If not, then there is stochastic independence between recall and recall.

If word lists are presented either under C1 or C2 and retrieved either under C1 or C2, will a double dissociation result, the cue at retrieval only facilitating recall if it matches the cue at encoding (by encoding specificity)? If so, there is crossover interaction within recall.

Independence and crossover interactions grossly overidentify dissociable systems. Unless *sufficient* conditions for dissociation are discovered within cognitive psychology, it will be necessary to validate putative system dissociation due to focal brain injury. Tulving appropriately emphasizes evidence for system dissociation derived from studies of the amnesic syndrome. The lesion disproportionately impairs memory for events (Kinsbourne & Wood 1982). Any cognitive system will, if it is real, sooner or later be selectively impaired in a case of focal cerebral damage. The experimental data on the intact case inform the neuropsychologist where to look.

Given the ample evidence for multiple systems in perception (e.g. Cowey 1979; Posner 1979), the assertion that memory is monolithic strains credulity. In any case, the opinions of commentators about whether there are one or more memory systems are mere opinion. As Tulving has pointed out, the necessary study has barely been begun. The question is, will we investigate cognition systematically, or continue to tinker with each other's paradigms as heretofore?

Variation on a theme: Are the elements of episodic memory dissociable?

Richard S. Lewis

Department of Neuropsychology, Neuropsychiatric Institute, UCLA, Los Angeles, Calif. 90024

Although there is general agreement about the heuristic value of the episodic/semantic distinction in memory, a question remains as to whether this involves functionally distinct memory systems. Recognizing this, Baddeley (1984), in his commentary on Tulving's (1984a) précis of *Elements of Episodic Memory*, focused on the neuropsychological evidence for a dissociation of memory functions corresponding to the episodic/semantic distinction.

Baddeley stated that the crucial neuropsychological evidence for the episodic/semantic distinction rests on whether amnesics can recall episodic memories. He cited two studies suggesting that amnesics can indeed recall prior events from their lives. Only one of these has been published at this time (Zola-Morgan, Cohen & Squire 1983). In that study, Zola-Morgan et al. reported that amnesic patients were able to recall specific autobiographical episodes in response to single-word cues. However, without validation that the amnesics' responses were in fact real events from their lives, this evidence against the episodic/semantic distinction remains questionable.

Furthermore, it may be misleading to evaluate neuropsychological evidence for or against the episodic/semantic distinction by whether or not episodic memory is merely present or absent. Rather than treating episodic memory as an indivisible memory component, we might find that neuropsychological evidence addressing the episodic/semantic distinction instead indicates whether or not *aspects* or "elements" of episodic memory are impaired relative to semantic memory.

Currently, there are at least two intensely debated issues in

the neuropsychological literature on memory (see, for example, Squire & Cohen 1984). One concerns the functional nature of the difference between impaired and preserved memory functions in amnesics and the other concerns whether different kinds of amnesic syndromes result from dysfunction in different brain regions. In Tulving's (1984b) reply to Baddeley, he suggests that it is important to keep in mind the second issue (i.e., that there may be different kinds of amnesic syndromes) in connection with the first (the episodic/semantic distinction). Tulving is right in pointing out that there is a tendency to treat these two issues as independent when, in fact, they may interact. He does little, however, by way of explaining how this interaction may occur. The purpose of this commentary is to suggest how these issues may interact and, to propose that different lesions resulting in amnesia may be affecting different aspects of episodic memory.

Squire and Cohen (1984) review evidence that "diencephalic amnesia" (e.g., in Korsakoff amnesics) differs from "medial temporal lobe amnesia" (e.g., the patient H. M.). In both types of amnesia the episodic/semantic distinction does a reasonable job of characterizing impaired and preserved aspects of memory (at least at a first approximation). This being the case, it is reasonable to suspect that damage in these different brain regions, sufficient to cause amnesia, may impair different aspects of episodic memory. One of the most distinct characteristics of episodic memory is the encoding of "focal elements" (focus of attention) with the "setting" or temporal-spatial context of the focal elements. In fact, successful retrieval of information from episodic memory tends to depend on using the temporal-spatial context to help retrieve focal elements. This differs from the retrieval of information from semantic memory, which tends to be context-free (Kinsbourne & Wood 1975). Could it be that diencephalic amnesia and medial temporal amnesia reflect impairments of different aspects of the contextual encoding or retrieval of episodic memory?

Damage to the mediodorsal thalamus (MD) appears to be sufficient to produce an amnesic syndrome (Speedie & Heilman 1983; Squire & Moore 1979; Victor, Adams & Collins 1971). MD is intimately connected with the prefrontal lobe (which can be defined as the cortical area receiving afferents from MD). Fuster (1980) has proposed that the function of the prefrontal lobe is to organize behavior temporally. The MD-prefrontal system, with its great number of reciprocal connections, may therefore be involved in the temporal organization or context of episodic memories. In support of this idea, Huppert & Piercy (1976) found Korsakoff amnesics to be impaired in judgments of temporal order on a recognition task. In contrast, medial temporal lobe amnesics have not been found to be impaired on temporal order judgments (Milner 1971; Sagar, Gabrielli, Sullivan & Corkin 1984; Squire, Nadel & Slater 1981). One exception to this pattern is the report by Squire, Nadel, and Slater (1981) in which N.A., a patient with a stab wound to the left MD region, did not differ from controls in temporal order judgments. This study has been criticized (Hirst 1982) for making comparisons between recognition and temporal order when accuracies were very low, possibly reflecting "floor" effects.

Another aspect of contextual encoding and retrieval of focal elements is the environmental context, or the entire stimulus situation surrounding a focal element. Considerable evidence exists that the hippocampal system is involved in processing spatial information (O'Keefe & Nadel 1978), an important aspect of the environmental context [see multiple book review *BBS* 2(4) 1979]. The hippocampal system appears, however, to be involved in much more than just spatial information processing (Olton, Becker & Handelmann 1979). Olton et al. proposed that the hippocampal system is involved in working memory, which both Olton et al. and Tulving agree is roughly synonymous with episodic memory. Olton et al.'s hypothesis is derived from lesion studies of the hippocampal system in rats, in which he finds that memories for food location in radial arm mazes are impaired. The rats appear to be impaired in their ability to use extra- and intra-maze cues to remember food location, that is, the environmental context (visual, tactile, spatial, etc.) of food location. This role of the hippocampus in the environment of episodic memories has been supported in other studies as well (Winocur & Gilbert 1984). There is, therefore, some indication that the hippocampal system is involved in the environmental context of episodic memories. Little research has been conducted studying environmental contextual processes in amnesics, and to my knowledge none with medial temporal lobe amnesics. However, Winocur and Kinsbourne (1978) have shown that Korsakoff amnesics can use the environmental contextual aspect of episodic memories.

There consequently appears to be a dissociation between temporal and environmental context in Korsakoff amnesics; they are impaired in using the temporal context of episodic memories but not the environmental context. Furthermore, it has been demonstrated that medial temporal lobe amnesics are not impaired in temporal context judgments, but no evidence exists as to whether they are impaired in using environmental context.

Other relevant evidence for the proposed hypothesis comes from the study of automatic processes. Hasher and Zacks (1979) proposed that frequency, spatial location and temporal order are automatic processes. Even though episodic memory is often considered to be an effortful process, the specific process of encoding temporal and environmental context along with focal elements tends to be automatic, and may therefore involve processes similar to those described by Hasher and Zacks. Consistent with the proposed hypothesis that the MD-prefrontal system is involved in the temporal context whereas the hippocampal system is involved in the environmental context, Smith and Milner (1983) found that frontal lobe patients were impaired in the judgment of temporal order whereas medial temporal lobe patients were impaired in the judgment of spatial location, one aspect of the environmental context.

Environmental and temporal context are not the only contexts that are used in episodic memories. Internal physiological and emotional states of individuals can also aid in the encoding and retrieval of memories, as witnessed, for example, in state dependent learning (see Eich 1980). Because the hypothalamic region is involved in the regulation of the internal environment, it may be that memory deficits found after damage to hypothalamic areas (Becker, Khan & Reddy 1984) are related to deficits in the amnesics' internal context of episodic memories.

This commentary has attempted to show that in discussing the nature of impaired and preserved memory functions in amnesics it is necessary to keep in mind that amnesia is not a unitary disorder. In addition, I have attempted to synthesize these two issues by proposing that amnesia resulting from damage to the hippocampal system and MD-prefrontal system reflect impairment of different "elements" of episodic memory.

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The crucial role of dissociations

Edward J. Shoben and Brian H. Ross

Department of Psychology, University of Illinois, Champaign, Ill. 61820

Central to any evaluation of the episodic/semantic distinction is one's evaluation of the merits of dissociation as an experimental strategy. Dissociation experiments have apparently provided critical evidence for some important theoretical distinctions in cognitive psychology (notably the distinction between short-term and long-term memory), but the diagnosticity of this evidence has been called into question both with regard to the short-term/long-term distinction (Tulving & Bower 1974) and

the episodic/semantic distinction (Hintzman 1984; Roediger 1984).

In the context of the episodic/semantic distinction, dissociation experiments typically show that a variable such as type of priming will have a marked effect on a task that is presumed to be episodic, for example, but little, if any, effect on a task that is presumed to be semantic. Tulving (1984r), for instance, cites a concatenation of experiments performed by McKoon and Ratcliff (1979) in which they found virtually no effect of type of priming on lexical decision latency but a substantial effect on recognition latency as evidence for the distinction.

In this section, we discuss criticisms of dissociation logic and consider how dissociations may best be used. In the next section, we evaluate other evidence Tulving (1984r) has proposed as supporting the episodic/semantic distinction. In the third section, we consider two possible modifications to dissociations, one suggested by Roediger (1984) and a new one based on the idea of higher order dissociations. Although there are difficulties with dissociation logic even after these modifications, it is our contention that this logic will prove critical in assessing the merits of the episodic/semantic distinction.

The primary criticism of using a dissociation to support a memory distinction is that any conclusion about two separate systems is premature. A theory postulating a single memory system may be able to account for an observed dissociation; because different processes are necessarily involved in performing different tasks, variables that affect one task may not affect another (e.g., Hintzman 1984). The classic example is the effect of word frequency on recognition and recall. High word frequency produces better recall but poorer recognition, yet no one has argued that these tasks tap different memory systems. On the contrary, there are several well-developed unitary memory theories (Anderson 1983; Kintsch 1980) that can account for this finding.

It is undoubtedly true that dissociations do not mandate the existence of two separate memory systems. On the other hand, the fact that they do not provide compelling evidence for two separate systems should not lead one to conclude that dissociations are not useful. What observed dissociations do provide is a constraint on theorizing that is often fairly severe. For example, any one who wishes to eschew the distinction between long-term memory and short-term memory must account for the differential impact of semantic and acoustic similarity on short- and long-term memory performance. Similarly, a good dual-process proposal must give a satisfactory account of why these dissociations occasionally fail to occur (see Crowder 1976, for a review).

Let us consider how these constraints have been applied to one line of argument concerning the episodic/semantic distinction. Shoben, Wescourt, and Smith (1978) observed a double dissociation in which semantic relatedness affected performance on a semantic task but did not affect performance on an episodic one; analogously, they found that fanning affected episodic performance but not semantic performance. Shoben et al. argued that their results supported the distinction and, in particular, were inconsistent with a prominent unitary theory, HAM (Anderson & Bower 1974).

Anderson and Ross (1980) worked generally within the constraint imposed by the Shoben et al. results. Although accepting the possibility that these findings were inconsistent with HAM, they argued that the results were consistent with a more modern unitary model, ACT (Anderson 1976). In addition, they also presented some data that they claimed were inconsistent with the episodic/semantic distinction.

It seems to us that this kind of sequence of events is a proper way for science to proceed. It is a shared assumption that theories that postulate only one memory system are preferable to those that require two or more; consequently, dual-system theorists must present evidence that they believe is inconsistent with unitary memory theory. Assuming some face validity for the results adduced, it is then up to the unitary memory theorists to assimilate the results into a unitary theory. In fact, researchers who have found many of the dissociations cited by Tulving have sought to achieve just this goal: Jacoby and Dallas (1981), for example, specifically reject the episodic/semantic distinction as an account of their observed dissociation, as do McKoon and Ratcliff (1979).

It would thus seem that any successful theory must account for the pattern of dissociations and failures to obtain dissociations. Tulving recognizes that any single dissociation can be explained by a unitary theory, but argues that if a separate explanation is needed for each dissociation then the unitary theory has no predictive power. Of course, the argument from the other perspective is that any two-system theory also needs to account for failures to find dissociations in a principled way. The constraints provided by dissociation experiments, then, derive not from explanations of single dissociations, but rather from accounts of the patterns of dissociations and parallel effects (i.e., nondissociations). Although all parties would probably concur with this summary, it seems there is a tendency for unitary theorists to point to the failures to find dissociations and a tendency for dual-system theorists to point to the dissociations.

One other point may be helpful in understanding how to account for both dissociation and parallel effects. In interpreting the results of dissociation experiments, one needs to consider whether the variable that produces the dissociation is peripheral or central to the task. The dissociation paradigm tacitly assumes that the manipulated variable affects a process that is central to the episodic/semantic distinction. As an extreme example of a peripheral process, the hypothetical observation that brightness level affected a visual lexical decision task but not an auditory recognition task would not be compelling evidence for the distinction. Although in the abstract it may seem easy to distinguish central from peripheral, in practice it is not. At least part of the argument over the merits of the evidence obtained from dissociation studies concerns whether the affected variable is affecting a central process or not. We ourselves have differed on this point (e.g., see the discussion in Anderson and Ross, 1980, about the Shoben et al., 1978, variables, and a critique of Anderson and Ross in Shoben, 1983).

The probity of the other evidence. Dissociation experiments are clearly not conclusive evidence for two memory systems; Tulving argues that they are a necessary but not a sufficient condition. He goes on to list all 11 other classes of evidence that he claims support the distinction. Unfortunately, it seems to us that these other forms of evidence are either very weak or subject to alternative interpretations. Consequently, we believe that results from dissociation studies must be the arena in which the battle over the merits of the distinction is fought. Given the importance we have attached to dissociations, it is incumbent upon us to demonstrate the weaknesses of the 11 other classes of evidence specifically mentioned by Tulving.

The first of these 11 cited by Tulving is armchair speculation. Although we agree that these kinds of ruminations are useful, they hardly constitute strong support; they merely provide some prima facie evidence that the theory is plausible.

Tulving's third point (the second is dissociation logic) is that recognition and fragment completion performance are stochastically independent (Tulving, Schacter & Stark 1982). Although stochastic independence within conditions does not depend on dissociations or lack of dissociations across conditions, the same problems of interpretation apply to it as to dissociations. On the one hand, stochastic independence may be the result of different systems. On the other hand, stochastic independence may occur because the information contained in the two types of probes is uncorrelated. (For an example of this applied to the near independence of recall and recognition, see Flexser and Tulving, 1978). At any rate, this particular line of evidence is questionable because Tulving et al. claim that the fragment

completion task is probably not a semantic memory task, but rather a task of a third, procedural system.

Many of the remaining lines of evidence cited by Tulving involve studies of the brain and amnesics. Although we admit the possibility that such studies may prove crucial in the future, they do not appear to be so at present. The studies showing types of brain differences are still very exploratory and are open to a variety of interpretations (see Baddeley 1984, for one example). The amnesic studies are problematic in that many of the results that purport to support the episodic/semantic distinction are really concerned with the procedural/declarative distinction. Baddeley (1984) has made this point very well. Although Tulving (1984r) does not agree totally with Baddeley, he does concede that much of the neurophysiological evidence is "not strong" (p. 260).

Tulving's ninth class of evidence is factor analysis, referring specifically to a study by Underwood, Boruch, and Malmi (1978). These authors were primarily concerned with the relationships among various types of verbal learning tasks, and their semantic tasks are not included in their factor analysis. Although Tulving is right that the episodic tasks generally show small correlations with the tasks characterized by Underwood et al. as semantic, the semantic tasks the latter used are far from typical. More specifically, Underwood et al.'s semantic tasks are: a comparative judgment of word frequency (Thorndike-Lorge), a spelling test, a vocabulary test, the SAT verbal and the SAT quantitative. Clearly, none of these tasks is one traditionally studied by semantic memory theorists. It would seem that most reflect intellectual ability and might themselves be little correlated with performance in a semantic memory task. As the term semantic memory is currently used, there is clearly nothing in the Underwood et al. study to make us believe that episodic and semantic memory are "different worlds."

The eleventh class of evidence is Tulving's claim that computers cannot have episodic memories. Although semantic information has been the primary concern of AI researchers, this past concentration may reflect their belief that semantic information processing was the locus of intelligent behavior. However, some recent work on learning and problem solving (e.g., Carbonell 1981; Kolodner 1983; Schank 1982) makes great use of episodic memories. Tulving seems to be claiming that machines cannot "experience" events, and so cannot have the richness of the episodic memory that people do. The argument is that computers are only "manipulating certain symbols according to certain rules" (Tulving 1983, p. 53), but such a claim is true whether computers are dealing with episodic information or with semantic information. [See also Searle: "Minds, Brains and Programs," *BBS* 3(3) 1980.]

Last, Tulving argues that the analogy to the visual system favors the episodic/semantic distinction. Here he claims that because the visual system is subdivided into a number of subsystems it is, therefore, likely that the memory system is similarly partitioned. It would seem to us that the likelihood of a similar partitioning might very well depend on the physiology of the two systems. Because so little is known about the physiology of the memory system, it seems premature to speculate about its similarity to the visual system. Despite this lack of knowledge, we would tend to agree that the memory system is probably subdivided into some more specialized structures. However, the question is whether one principal subdivision is the distinction between episodic and semantic memory. The analogy to the visual system provides little guidance on this question.

Our point here is not to argue that there is no valid distinction between episodic and semantic memory. Our reading of the evidence suggests that it is clearly inconclusive. The point of the preceding discussion is that the results of dissociation experiments are particularly important in part because the other bodies of knowledge are singularly unhelpful in assessing the verity of the episodic/semantic distinction. As we and others have also noted, the dissociation logic is also not as strong as one might have initially believed. The question then becomes whether some modifications can be made to the dissociation paradigm to make it more powerful.

Modifications to the dissociation paradigm. Roediger (1984) has already suggested one such modification. He argued that it would be a vast improvement if dissociation experiments used two tasks for each presumed memory system. One example of this might be to use both lexical decision time and perceptual identification as semantic tasks, and to use both recognition and recall as episodic ones. In this way one can determine whether dissociations occur across hypothesized memory systems or across tasks within a memory system.

This modification of the dissociation procedure still does not provide conclusive evidence, but it does provide greater constraints on subsequent theorizing. For example, it is considerably more difficult for a unitary theory to account for dissociations, because such a theory must now account for the performance pattern across four separate tasks. In a similar vein, a dissociation with one task but not another in the same system does not necessarily imply that the two-system theories are incorrect; however, such explanations require principled reasons why the dissociation occurs with one dependent variable and not with another.

It may be helpful to consider the logic and possible problems with Roediger's suggestion by using the ideas discussed earlier concerning central versus peripheral processes and patterns of dissociation. By increasing the number of tasks for which the dissociation must occur, Roediger is increasing the likelihood that the independent variable will be affecting a central process of these tasks. If the variable were not affecting a process central to the episodic/semantic difference then there would be no reason to believe that its effect on two episodic tasks should differ from that on two semantic tasks. (This reasoning also suggests the intuitive idea that differences between tasks should be maximized.) Although Roediger's technique may be more persuasive in finding evidence for the distinction, what does one conclude if an inconsistent pattern of dissociations is found? Because we have learned nothing about the nature of the processes being affected by the variable, we cannot be sure whether the lack of a consistent pattern is due to the falsity of the episodic/semantic distinction or to a poor choice of variables. But if one needs to "try" variables to see whether they lead to a consistent pattern, finding one is no longer so persuasive.

This discussion of Roediger's technique suggests that it might be fruitful to couple the dissociation logic with a methodology that gives further insight into the nature of the processes being affected. Although one's theoretical predisposition will affect the choice of methodology, we have decided to illustrate the idea of what we call higher order dissociations with the additive factors logic developed by Sternberg (1969). Obviously, this particular use of the higher order dissociation logic is thus limited by any of the limitations on the additive factors methodology (McClelland 1979; Sanders 1980; Taylor 1976).

To take a concrete example, let us consider a simplified version of the design used by McKoon and Ratcliff (1979). Subjects learn lists of words in pairs. They are subsequently tested on individual words where they must decide whether the presented string is a word (a semantic task) or whether the word was among those on the list (an episodic task). The independent variable is relatedness; target words are preceded on the list either by words that were paired with the targets on the study list or by words not on the list that are semantically related to the target word (such as lion-tiger). The critical dissociation is that the type of relatedness has no effect on lexical decision time but semantically related targets take longer to recognize than episodically related targets. This result is described in some detail in Tulving's (1983) book.

To add the additive factors logic to this task, we must first

decompose the lexical decision and recognition tasks into discrete stages. With simplicity a primary goal, let us assume that both tasks can be described by an encoding, a comparison, and a decision stage. Let us also assume that a variable like word frequency affects the comparison stage in both processes; it seems unlikely that frequency would affect any other stage. If these assumptions are correct, and if relatedness affects only the comparison stage of the recognition task, then we have satisfied the preconditions of additive factors logic. More specifically, we should find a different pattern of interaction in the two tasks. For the recognition task, we should find that relatedness and word frequency interact because they have a stage in common. For lexical decisions, however, we should find no interaction because the comparison stage that is affected by word frequency is not affected by relatedness. These effects must therefore be additive.

The benefit of the addition of the additive factors logic is that the locus of the dissociative effect is specified to a greater degree. In the hypothetical example outlined above, one can conclude that relatedness affects the comparison stage in an episodic task but not in a semantic task. Hence the task for a unitary theory is now to specify how the tasks differ such that the comparison processes in each might be subject differentially to the relatedness manipulation.

Thus, the higher order dissociation logic does not make dissociation experiments more definitive per se. However, it does seem to provide stronger constraints on both single-system and dual-system theories. The constraint can be even greater when one uses a more complicated higher order design. For example, the hypothetical example outlined above can be extended in a number of ways. First, we might want to include a variable like stimulus quality to ascertain whether the relatedness variable influences the encoding stage (Meyer, Schvaneveldt & Ruddy 1975). Second, we might want to increase the levels of relatedness (perhaps by including an unrelated condition) to determine whether any of the three relatedness conditions has an effect on lexical decision and, if so, on what stage. Finally, we might try to find a variable that affects lexical decisions but not recognition latency and see where this variable interacts with either word frequency or stimulus quality.

With these kinds of extensions it would seem that the results from these higher order dissociation experiments might place fairly strong constraints on any theory that deals with the episodic/semantic distinction. It is already difficult to account for dissociations from the perspective of a unitary theory, and it is similarly difficult to account for failures to obtain dissociations from the perspective of a dual-system theory. The results from higher order dissociations may make these tasks even more difficult in that they will necessitate a greater level of precision in our theories. If this analysis is correct then we will have achieved some success in addressing the complaint that it is difficult to assess the merits of theories that espouse the episodic/semantic distinction because they are so vague (Hintzman 1984).

Dissociation experiments, then, never provide definitive evidence for the existence of two memory systems. Particularly when the paradigm is modified, either by the multiple task approach suggested by Roediger (1984) or by the higher order dissociation methodology we have outlined, dissociations can provide important constraints on psychological theories and may even specify the locus of the similarities or dissimilarities. There is a danger that people may conclude that because dissociation experiments are not as conclusive as they were once thought to be, these experiments should be abandoned. If this abandonment takes place then we will have lost a useful tool in our goal of constructing precise theories of cognition.

Dichotomy: Strict or fuzzy

Ivan Šípoš and Jana Plichtová

Institute of Experimental Psychology, Slovak Academy of Sciences, 821 08 Bratislava, Czechoslovakia

Empirical dichotomies are products of observable or measurable features that are ordered by certain accepted criteria. Tulving's dichotomy of episodic and semantic memory concerns three basic diagnostic features: information, operations, and applications. For practical and theoretical reasons, these features are of the utmost importance and create the substance and the sense of a new approach to memory research. Observation and assessment of basic differences (Tulving 1972) preceded Tulving's later ideas of General Abstract Processing System and synergistic ecphory (Tulving 1984). The heuristic and theoretical value of this dichotomy is thus based mainly on arguments about differences. Human reasoning about dichotomies, however, seems to be contaminated by a bias toward discreteness, binary yes/no decisions. Cutoff points on some feature continua may be arbitrary, expressing either/or opposites where continuous quantity is more relevant. This type of reasoning has led to an explanation of the episodic/semantic dichotomy in terms of level of abstractness (Baddeley 1977). In the light of currently available evidence, the idea of a continuum is not appealing to Tulving (1984). There still remains the possibility, however, that Tulving's diagnostic features are not a strict limitation of both types of memory, that characteristic features of information, operations, and applications allow greater interdependence and interaction. We shall look more closely at these differences.

We may remember events that were not experienced perceptually and do not concern the self. These are episodes derived from symbolic operations, from reading, listening to other people, telecommunications, and so on. Famous historical episodes or even fictional events are often excellent anchors for remembering, for comprehension. Consider the episode about Newton's apple. Its source is neither sensory nor conceptual. It is an episode, temporally organized, not concerning the self. It is something like mediated experience, defined in time and space. Diagnostic features of information, operations, and applications are not univocally attributable to it, if at all. Some episodes of this sort have strong conceptual impacts. Over and above a simple image of a falling apple, there is the idea of gravitation. Ideas, concepts are anchored in a very concrete event. The case is very similar with figurative language, for example with proverbs. Two readings of a proverb are usually accepted: a literal and a figurative one. Proverbs may thus function both as episodes and as concepts that are prototypes of many episodes. Another example is dramatic art. Shakespeare's Hamlet consists of many episodes which all create an idea, and this idea changes across generations. It seems to be a fuzzy matter whether to characterize access to this as deliberate or automatic. Under the influence of stage fright, heavy fatigue or drugs, automatic access to semantic information may be disturbed, whereas pricks of conscience may be automatic companions of many painful episodes.

Tulving's summary of differences between episodic and semantic memory (Tulving 1984) represents, to our minds, the most important implications of his theory and the experimental data make sense only if this dichotomy holds. If the dichotomous features of episodic and semantic memory are fuzzy, however, then the whole theoretical structure seems to lose its force.

Some remarks are necessary regarding the principle of coding specificity. We are of the opinion that on the basis of this principle the validity of the following statements cannot be doubted: Some items (events) are remembered better than others: one coding operation is more efficient than another: a certain kind of prompt is more efficient than another, and so on.

Far be it from us to try to deny the validity of the principle of coding specificity. Nevertheless, no evidence has yet been provided that would rule out the possibility of some other principle being at work in the recall of a list of words. On the contrary, even Tulving's own results (Tulving & Osler 1968) suggest a possible interaction of coding specificity with the associativeness of prompts. Weak associations, for example, facilitate recall only if they are present at the time of both input and output, whereas strong associations facilitate recall even if they are presented at output only. Tulving's (1984) interpretations go only in the direction of specificity, which may entail some logical inconsistencies.

It has also often been confirmed that high-frequency items are better recalled than low-frequency ones (Bousfield, Steward & Cowan 1964), and that prompts with high associative strength are more efficient than prompts with low associative strength (Cohen 1963; Plichtová 1979).

Our remaining objections to the principle of coding specificity as a dominant principle are of a more general nature. Man specifically codes events, dependent on the context, so that he can discriminate them better; that is, he heightens the effectiveness of his memory. In this respect, experiments with coding specificity create a highly artificial situation. In addition, these experiments only concern the specificity of coding words. If we transfer our attention from words to the real elements of episodic memory, that is, personal experiences, then we cannot subscribe to the statement that all memory traces have equal strength. How is Lorenz's mechanism of imprinting to be explained, then?

Tulving (1984) considers it plausible that in child development, semantic memory precedes episodic memory. Many experiments, however, suggest the contrary. Thus, experiments with free associations suggest a marked developmental change from syntagmatic to paradigmatic associations. This shift is interpreted as a reconstruction of a cognitive thesaurus that is reoriented from episodic to systemic knowledge (e.g., Maršálová 1982).

Conscious constraints on episodic memory

Norman E. Spear

Department of Psychology, State University of New York, Binghamton, N.Y. 13901

I have read perhaps 90% of everything Tulving has published in the past 20-plus years. I have always been educated by his depth of analysis and wisdom, challenged by his argumentative nature, and delighted with the frequently "outrageous" style. I guessed that with *Elements* I might learn a few new things about memory and Tulving's approach to it but would not disagree with much. I was wrong on both guesses: I learned a very great deal and met a serious disagreement in the first line of Chapter 1.

The first two sentences in Chapter 1 read, "Remembering past events is a universally familiar experience. It is also a uniquely human one." No other animals can, says Tulving, "experience again . . . happenings from the past, and know that the experience refers to an event that occurred in another time and in another place" (p. 1). I do not disagree strongly with this assertion – who can know, one way or the other, what animals actually experience? – but I do dislike the narrowing of episodic memory. Memory data from nonhumans are of course excluded. I regard humans as highly as anyone, but there are too many experimental questions about memory that cannot be tested with humans and have been with animals, with important effect. The value of this effect should not be overshadowed by Tulving's acceptance that nonhuman animals "cannot travel back into the past in their own minds" (Tulving 1983, p. 1). This widely held belief (e.g., Kardos 1984) is not so much wrong in fact as in emphasis. There is reason to question whether human animals actually "travel back" in this way without continuous prompting by subtle events, external and internal, of the same sort that guide the memory of other animals. One need only glance at books on animal memory published within the past five years for varied illustrations of how a rat's, pigeon's, or monkey's memory for a discrete episode is degraded, revived, or changed in retrospect (or prospect) by minor events in brain or environment (e.g., Roitblat, Bever & Terrace 1984; Spear & Miller 1981). Although making no contact with the reinforcement contingencies that engage behavior in acquiring the memory originally, these minor events control behavioral changes that can only reflect the sorts of "scanning backward in time" that humans claim.

Most memory for events in humans seemed to me also excluded along with that for animals. There is of course nothing wrong with using the term "episodic memory" in Tulving's way (his invention; his priority) or in any other way, but to do so is neither to represent what has been studied about memory, as Tulving acknowledges, nor to identify a particularly interesting topic for the future. So, the spirit aroused in me was not so much adversarial as incredulous. Was Tulving really to devote an entire book to those cases of human retention in which individuals acknowledge verbally their awareness of having acquired the information requested of them at the memory test?

To me, instances of the "conscious-only" episodic memory have always seemed a small or at least unimportant fraction of what is learned. Tulving's eloquent insistence almost persuades me that I have been wrong. Yet, that learning from event contingencies occurs with no conscious acknowledgment may be verified so readily – with either real or "thought" experiments – as to require no examples. I have trouble assigning all such cases to the categories of semantic memory or procedural memory. And on the other side of the contrary case, how different are (a) a set of words one uses to describe one's awareness of the circumstances in which event combinations were learned, and (b) singular responses by infant humans or animals to the context in which the event combinations were learned? Not at all in function, if we exclude cases of the latter in which context acts merely as part of the conditioned stimulus.

In spite of the beginning, I proceeded undaunted to read every word of *Elements*. The book is in my opinion extremely valuable, indeed essential, for anyone who studies the behavioral manifestation of whatever is learned, by no matter what animal. Although depending ultimately on empirical facts for tests of his ideas, Tulving's style is quite unlike that of the "show me," dust-bowl empiricism of the midwestern United States, from which the study of human verbal learning and memory developed. He takes advantage, usefully I think, of "thought" experiments and compelling examples from our conventional experiences as rememberers. Instead of operational definitions he appeals to everyday examples to define important basic terms, such as the distinction between "setting" and "focal" elements (which refer to context and task elements, respectively) (pp. 143–44). This can be dangerous but I think Tulving applies it wisely. Tulving is sensitive to the reality that discoveries which really catch our eye are those that are counterintuitive with respect to our everyday memory. His "thought" experiments emphasize that we should not turn our backs on many thousands of years of day-to-day experience in learning and memory and reflections upon it. This brings to mind the charge given by David Krech to persons studying memory from a biological point of view: "Go constantly to look at memory in people. Know what you are studying, and whence came your question, and before whom your work will be judged" (1972, p. 223).

Tulving does not, however, surrender completely to the contemporary thrust of ecological validity (e.g., Neisser 1982). Tulving defends his study of memory in laboratory settings: "I know of no compelling reason why the general principles that apply to remembering of many events in the laboratory should be greatly different from those governing the remembering of

real life experiences. Rememberers do not leave their brains and minds behind, or switch them off, when they enter the memory laboratory" (p. 146).

Tulving prefers to view "encoding, recoding, and ecphory as more or less instantaneous changes in state rather than activities enduring in time" (p. 139). For Tulving these are more events than processes. This suggests a preference for macroanalysis and avoidance of the underlying details, which I find perfectly appropriate in our present state of analysis. Yet when detailed analysis is needed, as with the issue of recallable but unrecognizable words discussed in the last few chapters, Tulving applies it with diligence and effect. This represents an aspect of Tulving's work that has always impressed me – his capacity for depth of thought. It is characteristic of his writing that instead of 1 or 2 alternative explanations he typically has 5 or 6, and instead of 1 or 2 examples it is 4 or 5. In Table 2 of this book, instead of 3 or 4 differences of importance between episodic and semantic memory, he suggests 28. The elements of episodic memory discussed in Part II of the book and illustrated in Figure 7.1 are 13 in number. The uses of "retrieval" are not 2 or 3, but 8 and more (p. 177).

Another characteristic I like about Tulving's work is how unpredictable are his opinions and judgments – in some respects an operational definition of an interesting individual. I give three examples from this book of what I did not expect: (1) I suspected sometimes that a basic issue had been missed, only to discover that Tulving was leading up to precisely this point and with elaboration beyond anything anticipated. So it is that Tulving deals with overproliferation of memory systems. One alternative to introducing a new system for each new phenomenon is to introduce buffer concepts such as Tulving's "free radicals," which seem to me based on plausible notions of a knowledge system and address the development of a "semantic memory." (2) Equally unexpected was Tulving's treatment of the consequences of interference and repetition for learning, transfer, and remembering. He dismisses these (huge) issues in a few pages as a matter of recoding of the original memory (although he nearly acknowledges some limited success in the experimental analysis of such issues with paired associate learning) (p. 165–67). I happen to agree, fundamentally, with his ideas on these matters, but believe also that they warrant more elaboration. (3) I did not anticipate Tulving's mention of retrieval mode, aside from retrieval cues, as important for remembering. I think there is no such thing as retrieval mode. Differences in circumstances that might appear to influence promotion of retrieval are merely in what the rememberer is asked to express, explicitly sometimes but usually implicitly (outside of memory laboratories). Otherwise what goes on in retrieval would seem no different from what goes on in initial encoding or storage of the memory (Spear 1981; Spear & Mueller 1984). It is to Tulving's credit that in view of the lack of empirical evidence for anything like retrieval mode, he does not identify it as a formal element of retrieval within his system (GAPS).

I did expect that Tulving would be modest as to what psychologists have accomplished, but not so extremely. On page 129 he states that "cognitive psychology has not yet begun (the study of episodic memory)". Earlier he mentioned that the alternative system, semantic memory, has been studied less extensively than episodic memory and that practically nothing is known about procedural (skill) memory. This leaves us with the question as to what has been studied. Surely it has not been free radicals. Tulving in fact demonstrates in this book that a great deal has been learned about human memory, especially through his own discoveries. The descriptions of his own and related work are absolutely excellent – particularly the informal insights and revelations revealed in heavy black print.

Tulving's consciousness criterion for episodic memory warrants one more comment. Anyone familiar with the study of human "amnesics" during the past 10 years cannot help but be impressed by their characteristic tendency, at least with some etiologies, to exhibit acquired behaviors and yet explicitly deny any memory for having acquired them. Tulving's emphasis on this aspect of remembering indicates that he has been unduly influenced by this phenomenon. I, too, have used this evidence together with other data to argue that the "memory process" we are dealing with is by no means a unitary system (e.g., Spear 1984). Yet this effect in some cases of human amnesia, and with normal humans as well, is yet to be analyzed and is not understood. For neither amnesics nor normal humans is there information on, for instance, the base rate of false positives – the "awareness" of episodes that have not in fact occurred, as with the déjà vu sensation or perhaps some recall of dreams (e.g., Hintzman, Asher & Stern 1978). To decide about this would require an experimental design infrequently used in tests of human memory, although common in the testing of animals. This is a control condition in which subjects are given, during a "learning" stage, an episode different from the one tested for memory. In the absence of proper experimental analysis, I think Tulving has given this phenomenon disproportionate weight in the argument for a consciousness component in episodic memory.

Finally, two general comments about Tulving's book. (1) The content is undeniably biased and provocative, but, equally undeniably, it generates ideas. Experimental psychologists reading this book will be compelled to grab the nearest student or colleague as a subject to test Tulving's ideas against their own. (2) Scientists studying memory must develop a strange set of piecemeal reading habits. From titles and abstracts appearing in a wide variety of journals or edited books – from hard-core neuroscience to philosophical treatments of cognition – we arrive frequently at only one relevant article per journal issue or book. It is relatively rare that my professional reading involves a single book read cover to cover, and sometimes when it has, the redundancy has caused me to regret the time spent. Tulving's book is one that can be read cover to cover with ease, pleasure, and great profit. It is simply one of the two or three best and most important books of this kind I have ever read.

Author's Response

Episodic and semantic memory: Where should we go from here?

Endel Tulving

Department of Psychology, University of Toronto, Toronto, Ontario, Canada M5S1A1

The commentators have chosen to deal mainly with the issue of the distinction between episodic and semantic memory rather than with other topics discussed in *Elements of Episodic Memory* (Tulving 1983, henceforth referred to as *Elements*). Both explicit and implicit in their comments are useful suggestions for the improvement of the quality of the ongoing debate. This Response touches on four problem areas related to the episodic/semantic distinction that are discussed by the commentators: (a) classification of learning and memory, (b) systems and tasks, (c) dissociation and other kinds of evidence, and (d) episodic memory and consciousness.

Classification of learning and memory. The distinction between episodic and semantic memory can be considered to be a part of the broader issue of how we should think about classification in learning and memory. Classification is one of the primary activities of science; whole

branches of individual scientific disciplines are devoted to it. Although it has been a part of the broad field of learning and memory, so far it has proceeded on a relatively casual, unsystematic, and intuitive basis. Perhaps the time has come for students of learning and memory to start thinking seriously about the possibility of constructing a classificatory system that would describe and express the relations among the many varieties of learning and memory with which they are familiar.

Kinsbourne sees the distinction between episodic and semantic memory very much in this light, as a step toward what he calls "systematizing cognitive psychology." Implicit in his commentary is the call for the development of *systematics* as a new branch of the science of learning and memory, analogous to systematics in biology. We may imagine that, as in biology, systematics of learning and memory would entail three interrelated problem areas: *classification*, the ordering of varieties of learning and memory into natural groups; *taxonomy*, the study of the rules, principles, and bases of classifying; and *nomenclature*, concerned with the development and application of appropriate names for the varieties of learning and memory that the classification exercise would yield.

I completely agree not only with **Kinsbourne**'s general suggestion, but also with his recommendation that the organizing principle in the classification enterprise should be "the differentiation of neural systems subserving cognition," or what we might refer to as the *neuropsychological criterion*. I have discussed the broader classification problem in learning and memory elsewhere (Tulving 1985c).

Crowder's observation that as students of memory "we have been worrying too little about what it is we have been studying" can also be interpreted as pointing to the need for a natural classificatory system, as does **Spear**'s complaint against too narrow a definition of episodic memory in terms of mature human organisms and conscious experiences. A successful solution of the classification problem in learning and memory would help us understand what it is that we study in learning and memory experiments in a way not quite possible today. It might also illuminate the relation between people's ability to travel back in time in their own minds, and abilities corresponding to such episodic remembering in nonverbal and preverbal organisms, thus helping us to cope with problems raised by Spear.

Systems and tasks. A continuing source of difficulty in debating the distinction between episodic and semantic memory resides in the lack of agreement as to how we should think about the relation between different memory systems and different memory tasks. The problem is exemplified by a number of points raised by both **Crowder** and **Spear**.

In *Elements*, I distinguished between episodic and semantic tasks. Episodic tasks are those whose solution requires utilization of information acquired on a particular occasion, whereas semantic tasks can be solved on the basis of a person's general knowledge of the world alone. This distinction may be useful as a first approximation, but it fails to be a reliable guide to theoretical interpretations of experimental data.

I think it is wrong to identify particular tasks with, or "assign" them to, particular systems. Instead, we should accept the assumption that many typical laboratory tasks entail obligatory or optional component processes of all three major systems – procedural, semantic, and episodic. An important objective of research consists in a componential analysis of learning and memory tasks that would permit the assessment of the extent to which performance of any task depends on the three systems and their processes.

Acceptance of the assumption may lead to a research strategy somewhat different from the one that has been typically used to date. Instead of comparing performances on, and looking for dissociations between, "episodic" and "semantic" tasks, we might be better off comparing performances on tasks that are identical except for a single difference in a component process that we have reason to believe is required for one task and not another. To the extent that the critical component can be identified with a particular memory system, the outcome of the comparison would shed light on the properties of that system. For instance, if Task A calls for a simple judgment about the occurrence of an event in the rememberer's past and Task B calls for an additional judgment of the temporal date of that occurrence, we might want to assume that the episodic-memory requirements are greater in Task B than in Task A. An observed dissociation could then be considered relevant to the episodic/semantic distinction.

In working out the logic of such an enterprise we may have to begin with certain arbitrarily or intuitively derived assumptions, not only regarding the properties of different memory systems but also concerning the nature of the interrelation among systems. Right now I am partial to the idea – changed from the one inadequately formulated in *Elements* – that the relation between episodic, semantic, and procedural systems is "monohierarchical": Episodic memory is a single specialized subsystem of semantic memory, and semantic memory is a single specialized subsystem of procedural memory (Tulving 1984r, 1985a). This idea, like all others, will be modified or replaced when relevant empirical evidence so warrants.

As a single illustration of the problem of the relation between systems and tasks, consider the dissociation between episodic recognition and lexical decision expressed in terms of response latencies (e.g., McKoon & Ratcliff 1979). If response latencies reflect the operation of the procedural system – an idea suggested in *Elements* that **Crowder** is skeptical about – the dissociation implies that the procedural-system component of the episodic recognition task is different from the procedural-system component of the lexical decision task. Does such a state of affairs speak more to the procedural/declarative distinction than to the semantic/episodic distinction, as Crowder suggests? I think not. The fact that the procedural-memory component behaves differently depending on the nature of the required judgment about the "contents" of the memory store informs us of the differences in the judgment, or contents, or both, thereby implicating two kinds of declarative memory. But opinions of this sort leave the issue wide open.

This brief discourse on memory systems and memory tasks also implies an answer to **Crowder**'s question concerning the relation between serial learning and memory systems: Typical serial tasks in the verbal learning labora-

tory entail components of procedural, semantic, and episodic memories. The contribution that different systems make to the performance of different serial learning tasks, however, is unknown at this time, because the problem has not been systematically investigated. I have discussed the issue in a somewhat different context elsewhere (Tulving 1985d) and will not elaborate on it here.

Dissociations and other kinds of evidence. The distinction between episodic and semantic memory has attracted attention from individuals in different fields of research. Perhaps not surprisingly, the representatives of different scientific subcultures do not always see things from the same perspective. The commentators in the present instance nicely illustrate this state of affairs. **Shoben & Ross,** representing mainstream cognitive psychology, think that dissociation experiments play a central role in the task of evaluating the usefulness of the multisystem point of view. They offer constructive ideas as to how the usefulness of the dissociation experiment could be improved: by including more tasks in simultaneous comparisons (cf. Roediger 1984), by distinguishing between central and peripheral processes in tasks that show dissociation, and by combining the dissociation method with the logic of additive factors (Sternberg 1969). These are all welcome ideas from which the ongoing debate can only benefit. But **Shoben & Ross,** differing sharply from **Kinsbourne** and **Lewis,** give short shrift to other kinds of evidence that can be seen as supporting the distinction, mainly on the grounds that this evidence is "either very weak or subject to alternative interpretations." They are particularly skeptical of the usefulness of "studies of the brain and amnesics," although their misgivings are based on what they see as the current picture, rather than on what might happen in the future. I disagree with Shoben & Ross in this respect. I think that neuropsychological evidence will turn out to be crucial.

Lewis not only accepts the episodic/semantic distinction on neurological grounds, at least as a viable hypothesis, but also proceeds to refine the rules by which such evidence is brought to bear upon the issues. He reminds us once more that it is inappropriate to think of amnesia as a unitary disease entity and to base conclusions as to the reality of the episodic/semantic distinction on this assumption. I can only agree with most things that Lewis says. These include his criticism of the report by Zola-Morgan, Cohen, and Squire (1983) which Baddeley (1984) relied on in his own criticism of the episodic/semantic distinction. Just because amnesic patients can talk about their premorbid life events it does not necessarily follow that they are reporting the contents of their episodic recollective experiences. In some cases, at least, it seems rather clear that they do not (Cermak 1984; Cermak & O'Connor 1983). Here, too, our methods require refinement. The hypothesis advanced by Lewis regarding dissociation between the temporal context and the environmental context in amnesia is highly plausible. Since we have no reason to expect that knowledge concerning temporal dating and spatial locating of remembered events is subserved by the same brain mechanisms, selective impairment of these two component processes of episodic memory should be possible.

Episodic memory and consciousness. One of the most distinctive characteristics of "episodic remembering" is the rememberer's conscious awareness of the relation between his present recollective experience and the event from his personal past that the recollection represents. Elsewhere I have referred to the kind of consciousness that is necessary for episodic remembering as "autonoetic" consciousness (Tulving 1985b). Impairment, or loss, of autonoetic consciousness is characteristic of severe forms of amnesia in which patients can neither relive the past nor anticipate the future (Ingvar 1983; Tulving 1985b), although they can acquire some new knowledge.

Spear thinks that I have been "unduly influenced" by this phenomenon of the ability of amnesic patients to acquire new information without any conscious recollection of having done so. Spear's misgivings concerning the importance of the "consciousness criterion," because it is "yet to be analyzed and is not understood," contain a challenge to those, like myself, who think that autonoetic consciousness is the most fascinating phenomenon of the human mind. We must begin to take it seriously. It may be true, as Spear points out, that the "conscious-only" episodic memory constitutes but a small part of what organisms, including human beings, are capable of learning, or in fact do learn during their lifetimes. But what kind of criterion is the prevalence of a phenomenon in determining its value as an object of scientific study? Many important scientific discoveries or conceptual breakthroughs have revolved around phenomena that appear much more rarely in our everyday world than does consciousness.

In this context, I must also register my disagreement with **Shoben & Ross** who think that computers can make use of episodic memories. We know that computers do not possess anything like autonoetic consciousness. We know it with the same certainty with which we know – solipsists notwithstanding – that a real world exists "out there." If so, we also know that computers cannot remember their past.

Summary. The commentators have contributed constructively to the continuing debate concerning the distinction between episodic and semantic memory. They have pinpointed several problem areas in which further thought and study may shed light on the nature of the distinction and shape its future. These include the need for the development of a neuropsychologically meaningful classificatory system of learning and memory, a detailed logical and empirical analysis of the relations between memory systems and memory tasks, the working out of the dissociation logic and the means for evaluating the usefulness of other kinds of evidence, and the study of the relation between memory and consciousness. We have some ideas now about where we should go from here.

A further response. As the commentary by **Šípoš & Plichtová** was received after the others in the present round, it is dealt with separately. I briefly mention two agreements and three disagreements with these authors. Consider the agreements first.

It is indeed possible to think of episodic and semantic memories as constituting an experiential continuum, as I discussed in *Elements* (Tulving 1983 pp. 67–68) although

not in the *Précis* (Tulving 1984). And the revised formulation of the relation between the two systems (Tulving 1985a), which conceptualizes episodic memory as a subsystem of semantic memory, also makes it possible to think of the two as disparate regions on an evolutionary or a developmental scale. Nevertheless the hypothesis cannot be ruled out that the first continuum reflects the combined effects of two different systems and that the two regions on the second continuum are qualitatively different, in the sense that organisms with fully developed episodic systems are capable of memory feats not possible in organisms lacking episodic memory.

The second agreement concerns the issue of interdependence and interaction between episodic and semantic memory. Such interdependence and interaction have never been disputed. My 1972 essay already included a short discussion of the issue under the explicit heading of "Interdependence of the two systems" (Tulving 1972, p. 391). Today we know a bit more about the interdependence than we did in 1972, and we can raise more pertinent questions about it, as **Šípoš & Plichtová** have done.

Now the disagreements. I would not say that people *remember* historical or fictional events, certainly not in the sense in which they remember happenings from their own past. They may *know* these events as they know other things about the world they share with others, but such knowledge exists and can be used independently of episodic memory. When a person's episodic memory system is destroyed, as it is in certain cases of brain damage, the person may not be able to remember a single concrete event that he has personally witnessed or in which he has participated, although his knowledge of historical events is relatively intact (e.g., Cermak & O'Connor 1983; Damasio, Eslinger, Damasio, Van Hoesen & Cornell 1985).

My claim, both in *Elements* and in the *Précis*, that encoding specificity, or observed encoding/retrieval interactions, rules out *absolute* statements of certain kinds – e.g., that some items are remembered better than others, that some encoding operations are more effective than others, and so forth – does *not* imply, as **Šípoš & Plichtová** seem to think, that "all memory traces have equal strength." It implies that recollective experiences and memory performances always depend on synergistic ecphory, that is, on the joint effects of memory traces and retrieval cues. There are no good reasons to believe that the remembering of real-life events obeys different laws in this respect than does the remembering of miniature experimental events in the laboratory. Another comment that Šípoš & Plichtová make with respect to encoding specificity must also be questioned. It is true that strong associations in some situations facilitate recall even if they are presented only at retrieval, but it is equally true that in other situations they do not (Roediger & Payne 1983; Santa & Lamwers 1974; Thomson & Tulving 1970, Expt. 2). The encoding specificity hypothesis accommodates both of these outcomes, whereas some other explicitly stated theoretical ideas do not.

The third disagreement concerns **Šípoš & Plichtová**'s remark about the order of development of episodic and semantic memory. The issue is obviously important. It is also complex, and its resolution is likely to require a great deal of relevant evidence from many sources. The developmental shift from syntagmatic to paradigmatic associations may constitute a part of such evidence, but like many other observations in psychology, it can be interpreted in different ways. Thus, for instance, syntagmatic associations could be derived from children's memory for sentences they have heard, independently of any involvement of the episodic system, analogously with the phenomenon of source amnesia (Schacter, Harbluk & McLachlan 1984). The point is that mere ability to reproduce the contents of a perceptual input can not be regarded as adequate evidence for an intact episodic system, for reasons more fully discussed elsewhere (Schacter & Tulving 1982; Tulving 1983, pp. 29–31, 114).

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Commentary on **Robert J. Sternberg (1984) Toward a triarchic theory of human intelligence. BBS 7:269–315.**

Abstract of the original article: This article is a synopsis of a triarchic theory of human intelligence. The theory comprises three subtheories: a contextual subtheory, which relates intelligence to the external world of the individual; a componential subtheory, which relates intelligence to the individual's internal world; and a two-facet subtheory, which relates intelligence to both the external and internal worlds. The contextual subtheory defines intelligent behavior in terms of purposive adaptation to, shaping of,

and selection of real-world environments relevant to one's life. The normal course of intelligent functioning in the everyday world entails adaptation to the environment; when the environment does not fit one's values, aptitudes, or interests, one may attempt to shape the environment so as to achieve a better person–environment fit; when shaping fails, an attempt may be made to select a new environment that provides a better fit. The two-facet subtheory further constrains this definition by regarding as most relevant to the demonstration of intelligence contextually intelligent behavior that involves either adaptation to novelty, automatization of information processing, or both. Efficacious automatization of processing allows allocation of additional resources to the processing of novelty in the environment; conversely, efficacious adaptation to novelty allows automatization to occur earlier in one's experience with new tasks and situations. The componential subtheory specifies the mental mechanisms responsible for the learning, planning, execution, and evaluation of intelligent behavior. Metacomponents of intelligence control one's information processing and enable one to monitor and later evaluate it; performance components execute the plans constructed by the metacomponents; knowledge-acquisition components selectively encode and combine new information and selectively compare new information to old so as to allow new information to be learned.

Toward a synthetic approach to intelligence

Adam Biela

Catholic University of Lublin, 20-530 Lublin, Al. Raclawickie 14, Poland

The triarchic theory presented by Sternberg (1984b) is a synthetic and holistic approach to human intelligence that integrates the environmental approach and the cognitive tradition.

Since the first operationalization of the concept of "intelligence" by Binet and Simon (1905) there is still a strong tendency on the part of psychologists to treat intelligence as a disposition that determines one's school achievement or professional success. An academically and professionally oriented concept of "intelligence" is of much practical importance for making psychological predictions of success in a specific field but, unfortunately, it limits theoretical perspectives.

The environmental approach. Relevance and purposiveness define an intelligent behavior as a goal-oriented behavior in a given environment. This persepctive combines intelligence and the value system an individual takes from the social ecosystem in which he lives. Thus, intelligent behavior might be interpreted as wise in the sense that one's behavior fits the value system accepted in one's culture. Social acceptance should be emphasized here, otherwise we will have to qualify many socially unaccepted behaviors such as crimes as "intelligent."

An environmental approach allows one to understand intelligence as wisdom that relates to the preferred value system in a given culture. There are also some universal values, however, that transcend the specific culture and become human wisdom.

Intelligent behavior also has some connections with creativity. Although Sternberg's subtheory is not *explicit* about this connection, we can find some implications concerning creativity in the concept of "shaping" that deals with changing the environment in order to increase the fit between oneself and one's environment. However, creativity may also be related to adaptive capacities. The relationship between "intelligence" and "creativity" is more evident if we are considering the "stars" in any given field of endeavor. We can also speak about the "moral" or "social creativity" of individuals whose behavior is labeled heroic.

Facets of intelligence. The two-facet subtheory specifies the two following abilities: (1) dealing with task novelty, and (2) automatizing information processing. Sternberg presented two paradigms to measure the individual's skills in dealing with novel tasks. It is still hard, however, to define what this ability really means. I think it may deal with two different kinds of situations: (1) relatively new situations which have never been faced before (completely new phenomena, tasks); (2) known events seen in a new perspective. This is certainly multidimensional in its nature. The way to discover these dimensions is, I think, to examine various kinds of analogical inference situations that deal with novelty. Analogy tests should of course be treated as one possible research paradigm. Essential for "novelty ability" are the components that enable a person to grasp the "base relation" for analogy between the domains compared (see Biela 1981).

Components of intelligence. The specification of the components of intelligence depends on the level of analysis and the kind of problem being solved. Let us try to specify, for example, the components by which intelligent performance is generated in decision-making situations. We will interpret Biela, Chlewinski, and Walesa's (1983) proposal concerning developmental aspects of decision making in terms of Sternberg's componential approach.

First, the metacomponents have to specify the decision problem which needs to be solved: "Which of the alternative courses of action is capable of fulfilling the decision maker's expectation in an optimal or satisfying way under particular motivational and environmental conditions" (Biela, Chlewinski & Walesa 1983, p. 229). The decision maker may be aware of the decisional character of this problem in varying degrees. The result of this awareness can be viewed as a structuring process, a cognitive representation of the decision problem.

In accordance with the definition of decisions as sequences of processes we can specify at least the following five components of intelligent decision making: (1) the ability to perceive alternative actions in a given situation; (2) the ability to perceive the relevant states of the world viewed in the context of the alternative actions (i.e., outcomes of actions); (3) the ability to assess the probability of events that depend on alternative actions; (4) the ability to estimate the utility of actions relative to their outcomes; (5) the ability to seek and gather information suitable for the decision situation (see Biela 1976).

The components listed here are rather global units which may be differentiated into more detailed subcomponents, for example, the fifth component may be interpreted in terms of Sternberg's knowledge-acquisition components.

Concluding remarks. The three subtheories of human intelligence seem to touch on the important factors; however, more serious effort ought to be made to integrate them in one synthetic theory. The first step should be a synthetic definition of intelligence that would combine the subtheories discussed. I propose the following synthetic approach in defining intelligence: Human intelligence is an integrative disposition which organizes a person's specific abilities (componential subtheory) to perform in a new or familiar kind of situation (two-facet subtheory) in order to achieve desired goals in a given environment (contextual subtheory). This integration seems to be central to the concept of intelligence. The appropriate neuropsychological foundations for this disposition may reside in the theory of the integrative activity of the brain (see Konorski 1967). A complete theory of human intelligence must encompass three domains: (1) biological foundations, (2) cognitive processes, and (3) the axiological level.

The axiological level contains the value system a person takes from the society in which he lives. If psychological theories do not imply this level they are still far from being a complete explanation of human intelligence.

It is worth stressing that the concept of value- and goal-oriented intelligence was proposed by Stern (1920) and by Szuman (1931) but, unfortunately, this line of thinking has not been followed up in psychological theory. However, we have

some empirical arguments that the cognitive level should not be separated from the moral level. From the studies of Piaget (1948) and Kohlberg (1969) we learn that higher levels of moral development are related to higher levels of cognitive development.

I have tried to suggest in this commentary that it is possible to include the axiological level in Sternberg's triarchic theory. This level would combine the social environment from which a person takes his value system and the values themselves. An "axiological supplement" to a theory of human intelligence will not decrease its empirical testability.

Theory? Or tools for social selection?

K. Richardson

School of Education, The Open University, Milton Keynes MK7 6AA, England

Sternberg's (1984b) article represents a massive effort toward progress in the area of intelligence, not least because it helps draw a wide range of issues into focus. It is not a coherent theory, however, which I would take to mean a clear "model" which makes predictions concerning, for instance, the consequences of certain circumstances or of specific human interventions. Rather it is a framework for particular research programmes which are, here, being very strongly advocated. Such frameworks can be very useful so long as they keep the goal of theory in mind, lay down the criteria of such theory, and promote research accordingly.

Unfortunately I suspect that the goal of Sternberg's framework appears to be primarily to furnish tools of social selection, and only secondarily a scientific theory of intelligence. Because of this it appears to be fatefully corrupted by the old "strength" model of intelligence. This is immediately apparent from the way in which traditional metaphors like power, strength, weakness, brightness, and so on, crop up in the text as fundamental conceptions of intelligence. Sadly, what this means to me is that the enormous research programs that Sternberg is seeking to stimulate will turn out to be as awry as those we have seen for most of this century. Although I think that the way Sternberg chooses to cleave the area into three subtheories is itself unfortunate (and there are many inconsistencies in the text resulting from that cleavage), perhaps I can best make my point by addressing my comments under Sternberg's three main headings, although there is inevitably much overlap. I will have more to say about a framework for theorising about intelligence in my concluding comments.

The contextualist subtheory. It is important to point out, as Sternberg does, that intelligence is relativistic and personally historical: that what constitutes an intelligent act in one social situation may not be in another, and so on. But beyond this I find the contextual subtheory disappointingly narrow and superficial. The two problems that stick out most are, first, the omission of any allusion to the ecology and evolution of human intelligence and, second, a far too narrow view of the way in which humans encounter and solve problems in context.

Certainly much has been written about the circumstances of human evolution, and if we expect (as we surely must) that the ecological context might have determined the shape of that evolution, and with it the nature of human intelligence, then ecology ought to be part of any "contextual" framework for theorising about intelligence. The consequences of human evolution are, in fact, inescapable: Although Sternberg claims to be addressing only individual intelligence, all the "contexts" he cites – in which the properties of adaptedness, relevance, shaping, and so on, are manifested – are social contexts. This will not be the least bit puzzling to those (e.g., Holloway 1975; Jerison 1973; Washburn & Avis 1958) who have pointed out that social cooperation was the context of human brain evolution, including its rapid tripling in size, and therefore most probably had something to do with the fashioning of human intelligence.

This point is ironic, in view of Sternberg's stress on novelty, because the advantage of a social-cooperative mode of adaptation in evolution was precisely that, through a "shared" intelligence, stemming from joint-perceptions and joint-actions (see e.g., Leakey 1979), our hominid ancestors could deal with the extreme degrees of ecological change and novelty with which they were faced. What this means is that dealing with novelty is not only of the essence of human intelligence, and the product of severe selection pressures, but also that the group is its primary locus. The question is, therefore, to what extent *can* individual and group intelligence be meaningfully separated?

The other, and related, danger is that of characterising contexts (and with it the makeup of intelligence) too superficially. Sternberg seeks to define contexts in terms of "tasks"; and to define intelligence, corresponding with this, as "the potential set of *contents*" that count as intelligent. This aspiration is, of course, part of Sternberg's goal of tools for social selection. But I think it *is* a superficial view, which overlooks the fact that ecological and social contexts are characterised, not by isolated "tasks," but rather by the deeper "structure" by which objects and events are related to each other. Without that deeper structure there would be no context, only chaos, and no organism would be able to adapt to it (Colwell 1974; Plotkin & Odling-Smee 1979). It is thus incumbent on the adaptive machinery (in this case intelligence) to somehow abstract that structure, before it can even identify specific tasks. That is, "tasks" aren't things which simply crop up in contexts. It is the human intelligence, with its in-depth representation of that context which derives and defines them. Thus, I think the idea of a one-to-one "fit" between specific tasks and contents is a rather superficial one and that Sternberg's whole picture of the way in which humans encounter and solve problems in contexts needs to be questioned.

The two facet subtheory. In this aspect of his framework Sternberg claims that intelligence is *most* involved in the continuum of a person's experience where there is most novelty; and that it *becomes* involved by the recruitment of a process of "automatization."

Novelty. It is right that Sternberg should stress the importance of novelty in any model of intelligence. But he probably misses the point when he emphasises novelty "outside people's ordinary experience." For novelty, and environmental change generally, are very much part of the ordinary experience of people. Almost everything we see, for instance, is novel in the sense that we are always seeing things from different angles, orientations, and so on, or in the sense that few natural objects or events are exactly alike. And how we are capable of generating an infinite variety of novel responses to deal with this common experience is, as Hebb (1949) put it, "a fundamental ubiquitous psychological problem" (p. 153). If automatization is the solution, then it implies a very special use of the term which Sternberg does not disclose.

In psychology the problem has been at the heart of debates between rationalists, empiricists, and (since Kant) constructivists for hundreds of years. It has been the constant bone of convention about learning and the nature of knowledge. Dealing with novelty has always been assumed to involve some kind of concept formation. Sternberg writes that "intelligence involves . . . the ability to learn and reason with new kinds of concepts." But concept learning and usage is a deeply problematic area. It is an area that has attracted a good deal of research lately, but the literature shows that it is still very poorly understood (Medin & Smith 1984; Smith & Medin 1981). I am surprised that Sternberg does not bring this long-standing tradition of psychology to bear on his discussion of novelty. Theory based on premises which are themselves divided and confused is unlikely to be fruitful.

Automatizing. Sternberg's model of automatization is in itself a dualistic one that is reminiscent of some put forward in the

Middle Ages. I wonder whether the recourse to a central executive or homunculus serves the same purpose as the soul did for Descartes and his followers, namely, to escape the horror of reducing a person to an automaton, which would otherwise be the case? Many who have commented on Descartes' dualism have sought its roots in the growing individualism in all spheres – philosophical, social, economic – at the time. The problem for Descartes and his followers was, as Whitehead (1932, p. 179) put it, to make "minds and bodies exist in such a way as to stand in need of nothing beyond themselves individually."

Thus I wonder whether Sternberg's dualism is also a consequence of isolating the individual as the locus of intelligence, and of the need to escape the spectre of the automaton that immediately springs up. Of course, it corresponds nicely with the distinction between "entrenched" and "nonentrenched" tasks, but I've already mentioned my worries about Sternberg's picture of the way we encounter problems. Therefore, I have many doubts about it as an adequate description of the difference between a novice and an expert. I may be a better driver than I used to be, but I hope this *doesn't* mean that I devote less "attentional resources" to my driving and conduct myself on the road like an automaton. Similarly in the laboratory, lecture room, home, garden, and any other domain of expertise I can think of. What I vaguely feel I *do* have is more complex conceptual representations of situations – in which novelty is a *constant* factor – and more complex "motor schemata" (Schmidt 1975) for dealing with them. When I do move into a completely new domain, let us say Japanese cookery, with new implements, materials, and procedures, this doesn't present itself to me as an individual novel "task" that my individual "ability to deal with novelty" has to solve. Because, as human beings, many of our concepts or schemata are already shared, I can relate those of an expert (in the form, say, of a recipe book) to those I already have, and extend my "intelligence" accordingly. In sum, I have doubts about the dualism because I have doubts about the individualism on which it may be based.

Even if we *were* to accept the dualism, the executive and the automatization, there is no indication in this subtheory how they actually work. The problem of dealing "routinely" with novelty is not a trivial one. It has preoccupied philosopher-psychologists for 2,000 years, and the alternatives proposed – innate ideas, associationist abstractions, and Kantian constructions – are still before us, pretty much unresolved. Again it is no use pretending that unresolved matters like this *have* been resolved.

The componential subtheory. Just as we have all been warned about the nominal fallacy, I suspect that there may be a "componential fallacy" rapidly developing in cognitive psychology. This is the belief that an impressionistic breakdown of an activity into smaller and smaller steps is characterising processes actually occurring in nature (in this case in the human brain). The danger is not only that our superficial impressions might be wrong, but that they may lead us further away from reality. For instance, it is possible to describe a componential theory of memory. But the remarkable uniformity of structure, multiple representation, and interconnectedness of the cerebral cortex could easily confound this. As Edelman (1978, p. 71) put it, "Memory is not a localised property of any particular region of the nervous system . . . it is a *general* reflection of the enhanced interaction of all groups . . . there is no need to posit memory as a property of a given region or as an exclusive emergent property." Likewise I think the fractionation of intelligence into increasing numbers of separate components will only get us farther away from what we most need, which is a model of a unified system that carries out complex functions simultaneously within the brain as we currently know it.

Another danger is that a purely formal fractionation may in fact produce a delusion of further understanding where none, in fact, exists. Thus Sternberg has "identified" (imagined?) a "decision" component that allows us to "first figure out just what the nature of the problem is." (Might we also call this a "classifying" component?) There is then a "selection" component which picks out the appropriate "lower-order components." The appropriate "performance components" will then solve the "task" by encoding, combining, and comparing stimuli, and issuing responses. So, I might be confronted with a problem; my homunculus decides "ah ha! this is an inductive reasoning problem" and passes it on to the inferring relations component, which duly encodes the stimuli, compares or combines them (with the help, presumably, of representation or organisation components) and issues a response. I hope this is not too much of a parody of what is obviously an enormous effort. But my problem is that my understanding of intelligence is increased not one jot by this kind of account (quite apart from the host of fundamental questions that go begging within it), and the processes themselves remain as locked away behind their seductive labels as they have ever been.

An even more general problem, however, is that in addition to not explaining how these components work cognitively and cerebrally, the theory does not explain where they come from, or how they develop, in the individual, or how they change historically either in the individual or in society, in, say, the change from a preliterate to a literate culture, or from handcraft to mechanical production. To suggest, as Sternberg does, that all this might be explained merely by the use of different permutations of universal components begs more questions than it answers, to my mind. It does not explain the enormous creativity of humans in history; nor does it allow for the possibility of induction of entirely new routines or "repertoires" that are now hypothesised purely on the basis of recent developments in neurophysiology (Edelman 1978).

Conclusion. Sternberg's contextual subtheory is probably too narrow because it ignores ecological, evolutionary, and sociohistorical roots of human intelligence and, in simply juxtaposing "tasks" with "skills" or "abilities," misunderstands the way humans encounter and solve problems in context; his two-facet subtheory overlooks problems raised in classic treatments of novelty, especially in recent debates about concept learning; and his "dualistic" apposition of executive and automatizers is possibly only made necessary by the attempt to isolate individual from social intelligence. Finally the componential subtheory could be entirely fallacious, producing a delusion of further understanding where none exists.

I think most of these problems stem from Sternberg's framework, which subordinates scientific theory to the social goal of devising tools for social selection. Though obviously a bold effort to go "beyond IQ," I suspect it will need to be watched carefully lest the research programs it advocates fall into the same old traps. This is an area in which psychologists have no greater responsibility. IQ theory has seriously affected the lives of countless millions of people in the twentieth century, and it is no joke that, largely because of IQ theory, a substantial proportion of people, perhaps a majority, have left, and are still leaving, school throughout the world, convinced they are incapable of learning anything very serious.

Anything which improves that appalling situation must be welcomed. But I think that will need a framework that puts the priorities the other way round, and subordinates tools for social selection to the development of detached theory. Such theory should aim for an interdisciplinary description of human intelligence as a complex psychobiological system; relates intelligence to its ecology, evolution, and social history; at least attempts to relate it to the current knowns of higher cerebral functions; and embodies laws of its development, in the individual and social sense, in a way which can predict the outcomes of certain experiences, and of certain human interventions, and in particular state the minimal conditions of its liberation and fruition in all humankind. Only then can we pretend that our tools of social selection are based on firm theory, rather than

interim strategems. This, of course, presents a scientific problem that far outstrips in complexity and magnitude anything that any other science has had to face. We must at least thank Sternberg for giving the area an enormous stimulus.

Ghosts of the homunculus and of Sigmund Freud

Herman H. Spitz

E. R. Johnstone Training & Research Center, Bordentown, N.J. 08505

Whether or not one agrees with every (or any) aspect of the triarchic theory, we owe a debt to Sternberg (1984b) for organizing a large amount of data and many diverse theoretical viewpoints into a system that will surely prove useful in many ways, some of which were pointed out by the commentators. There is, however, one section of his presentation that was uncharacteristically illogical. I refer to the discussion of the executive process that is said to direct the nonexecutive processes to carry out the strategies it selects, monitors, and revises (pp. 277–78). The notion of an "executive process" in the brain is always an invitation to fuzzy thinking. One must wonder, for example, how a *process* can direct anything, no less another process. But then the executive is not always a process, sometimes it's a thing ("A central executive directly activates nonexecutive processes" p. 278) that is capable of "recognizing" potentially relevant situations. Still later it is the entire person who selects strategies for combining lower-order components (e.g., "One must also sequence these components in a way that facilitates task performance" p. 282).

If, for the phrase, "A central executive activates a particular production system," we substitute the phrase, "Joe Smith activates a particular production system," nothing seems to be lost except the illusion that we had somehow increased our understanding by transferring a function from the unknowable mind to an invisible homunculus residing in the mind.

There are so many good things that can be said about Sternberg's paper that this bit of carping may seem unfair, but it is only because the theory is so important that its inconsistencies should be pointed out.

In a lighter vein, it also occurred to me that the theory occasionally parallels in interesting ways some aspects of Freud's triarchic theory of emotions. There is, first, the superego, the individual's incorporation of society's customs that are communicated via the parents, that is paralleled in Sternberg by aspects of the contextual subtheory that "relates intelligence to the external world of the individual" (p. 269). There is the id, made up of innate, universal urges, and mirrorred in the componential subtheory composed of mental mechanisms that are "the same across all individuals and sociocultural settings" (p. 270). Finally, there is the ego, which moderates between the expressive urges of the id and the repressive forces of the superego, corresponding to that aspect of the two-facet subtheory "which relates intelligence to both the external and the internal worlds of the individual" (p. 285).

In Freudian theory, too, the universal features can be expressed differently in different cultures, but I agree with Sternberg that universals must exist. After all, we all have human central nervous systems and therefore are constrained to think in certain fundamental ways. Indeed, if one searches carefully one can always find instances of invariant intellectual behavior hidden in outwardly diverse cultures throughout the world (e.g., Spitz 1978).

Author's Response

Alternatives to the triarchic theory of intelligence

Robert J. Sternberg

Department of Psychology, Yale University, New Haven, Conn. 06520

The critiques by **Biela, Richardson** and **Spitz** merit three completely different responses, and hence I will reply to each separately.

"Yes": A reply to Biela. In my article (Sternberg 1984b) and especially in my subsequent book (Sternberg 1985), I show how the triarchic theory of human intelligence evolved from previous theories that have tended to dwell on just one of three questions: (a) What is the relation of intelligence to the internal world of the individual? (b) What is the relation of intelligence to the external world of the individual? (c) What is the relation of intelligence to experience? I argue that a fairly comprehensive theory of intelligence should address all three of these questions, not just one or even two of them.

The triarchic theory is not the only possible theory that might address all three of these questions, and indeed, I suspect that a major direction that future theorizing about intelligence will take will be to answer all three of these questions simultaneously, but in different ways. Biela's response represents an attempt to do so. We are in agreement as to the questions to ask, but Biela answers the questions somewhat differently (although with substantial overlap with the triarchic theory). Of course, Biela can sketch out his own constructive program of theory only briefly in his reply, but I believe his alternative formulation, if elaborated, would provide an engaging alternative to the triarchic one.

I believe that the triarchic theory is no less integrated than is Biela's theory, although the full extent of the integration may be more obvious in the book (Sternberg 1985) than in the article (Sternberg 1984b). Nevertheless, I welcome Biela's critique as a constructive attempt to explore alternative formulations of intelligence within what I, at least, perceive as a triarchic framework. As Biela shows, such a framework allows a multiplicity of theories, selection among which must await future research.

"No": A reply to Spitz. I respect, but disagree with, the bulk of Spitz's critique.

First, Spitz obviously does not welcome the thrust in contemporary cognitive psychology to separate executive from nonexecutive processes, or cognition from metacognition. Although the model seems to bother Spitz, I am not clear as to why. If one views intelligence as mental self-government, which I do, then it might be useful to draw an analogy to government of political bodies. An executive (a person, say, Ronald Reagan) fills a role (or the executive, say, the presidency) and has primary responsibilities (executive processes) that in turn provide direction for lower functionaries in government. The processing of information by these lower functionaries (presidential staff) is very much affected by the processing of information by executive orders, just as the executive's

orders are influenced by feedback from subordinates. I draw this analogy merely to point out that the logic of government applied to political entities is often carried over to psychological entities in contemporary cognitive psychology. Analogies, by their nature, are imperfect. It would be foolish to claim that the mind works just like a political government, or a computer, or a telephone switchboard, or like whatever metaphor one prefers. But metaphors always have been useful in scientific explanation, whether in psychology, biology, or physics, and I believe that they are useful in helping us to conceptualize intelligence. Any metaphor can be carried to the point of absurdity – for example, when one looks for silicon chips or computer programs in the brain – but I believe it is Spitz, and not I, who pushes the metaphor beyond the point of usefulness.

Second, there is Spitz's own analogy – between my own theory and Freud's. I will have to leave the evaluation of this analogy to the reader. If Spitz has trouble with my analogy, than I perhaps have double-trouble with his: I can't even get past the inference component!

"You must be kidding": A reply to Richardson. Richardson refers to my framework as furnishing primarily "tools of social selection" and only secondarily a "scientific theory" of intelligence. He never explains why he believes this, and in the absence of either documentation or elaboration, I take the point to be primarily contentious but of unknown and possibly no substance.

Richardson also refers to my theory as "fatefully corrupted by the old 'strength' model of intelligence," by which he means that individual differences in level or quality of task performance are dealt with in the text. Richardson's point is correct, but so what? In the first place, my theory is not *derived from*, but rather *accounts for*, individual differences (see especially discussions in Sternberg 1977; 1985). In this respect, it differs from factor-analytic theories. Second, a good theory of intelligence, or of any other psychological construct, *should* be able to account for individual differences. Third, Richardson's own conceptualization is no more immune to his own critique than is mine. He suggests psychologists use concept learning as an entree to the study of intelligence, and yet, even the earliest studies of concept learning demonstrated individual differences in it (e.g., Bruner, Goodnow & Austin 1956), both in terms of levels (quantitative differences) and styles (qualitative differences).

The fact that I choose to deal with individual intelligence does not preclude my dealing with it in social context. Indeed, that is what the contextual subtheory is all about! Here, Richardson seems to be attacking a straw man.

I also agree with Richardson's emphasis on the importance of understanding intelligence in an evolutionary perspective. I also believe it important to understand the neurophysiology of intelligence as well as its partially genetic basis. But no one theory of intelligence, at least at the present time, is likely to deal with intelligence in *all* of its aspects. Certainly, no theory has yet done so. I think it fair to say that the triarchic theory is among the broadest theories of intelligence that have been proposed – some find it too broad – and if there is yet more to be dealt with, I, and I hope, others, can be willing to accept that as a task for the future. No theory is ever complete, and the triarchic theory, of course, is no exception to this generalization. I hope, similarly, that Richardson will acknowledge that a conceptualization of intelligence in terms of concept learning might itself be less than complete.

I agree that the characterization of contexts in the triarchic theory is woefully inadequate. Providing a theory of situations has proved to be extremely difficult, not only for myself, and I welcome attempts by contextualists such as Berry (1984) to provide such theories.

Richardson's comments on the two-facet theory are quickly answerable. Of course, novelty is part of most people's "ordinary experience." But to avoid self-contradition, Richardson uses the term *ordinary* at a lower level of abstraction than he uses the concept of *novel* (i.e., ordinary as opposed to extraordinary novelty). When the terms are used at different levels of abstraction, their use is consistent with my own. When they are used at the same level of abstraction, they are contradictory. I am not sure wherein the disagreement lies. Richardson's remark that my discussion of automatization raises "the spectre of the automaton" is about at the level of the statement that cognitive psychologists turn people into robots. This kind of commentary may have appeal to the masses who read the popular press, but seems below the sophistication of the readers of *BBS* with respect to current paradigms in the study of cognition.

Finally, I do not believe that my componential subtheory is likely to "lead us further away from reality." Hundreds of cognitive psychologists, myself included, have done experiments in which they have decomposed performance on cognitive tasks into relatively elementary components (see, e.g., Sternberg 1982; 1984b). These studies have tested both the internal and the external validity of these componential decompositions, in many cases with highly favorable results. Before the componential approach was tried, Richardson's point of view was shared by many. I would hope that those who have read the cognitive literature would share it no longer.

To conclude, I welcome these and other critiques of the triarchic theory of human intelligence. The triarchic theory, like all other theories, is transitional, and critiques such as these help move us along to the next steps in theorizing.

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